Log-concavity criteria for finite nonnegative linearization

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Abstract. In a 2005 survey, R. Szwarc asked for criteria, stated in the recurrence coefficients, for nonnegative linearization (property (P)) of finite orthogonal polynomial systems such as the Krawtchouk polynomials. For a finite three-term recurrence normalized to have row sums one, write $d_i = b_i + c_i$ for the profile and $\rho(i) = c_i/d_i$ for the ratio of the downward coefficients to it. We prove that property (P) holds for every product, including products beyond the terminal degree, whenever the Newton coefficients of the profile are log-concave without internal zeros and the ratio is constant. The ratio may also increase with the index: if in addition the Newton coefficients of the ratio are nonnegative and log-concave without internal zeros and the diagonal coefficients b_i are nondecreasing, property (P) still holds. Separate theorems admit all convex quadratic ratios and, under the single inequality $r\sigma \le \frac{7}{4}t\tau$, the cubic ratios $r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The constant-ratio criterion contains the full positive Krawtchouk range and the dual q-Krawtchouk arrays, and through a quadratic decomposition of symmetric finite systems it covers Szwarc's quadratic system Q_n in its sharp range $0 < p_0 \le 1/2$. As a limit to any such criterion, property (P) fails at coefficient distance $O((1-r^2)^N)$ from the Krawtchouk array of size N with fixed ratio $r < 1$. The key inequalities of the increasing-ratio criteria are proved with exact computer certificates, and all theorems, corollaries and propositions are formalized in Lean 4.

Keywords. Finite orthogonal polynomials; nonnegative linearization; Krawtchouk polynomials; log-concave sequences; Newton series; quadratic transformation; polynomial hypergroups; computer-assisted proof.

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1. Introduction

Nonnegative linearization (property (P)) asks whether the product of two orthogonal polynomials is a nonnegative linear combination of polynomials of the same family. For a finite system $R_0, \dots, R_N$ the identity is understood on the orthogonality support, so it includes products whose degree sum exceeds N. In the normalization $R_i(1) = 1$ the coefficients then sum to one and define a finite polynomial hypergroup.

In his survey, Szwarc [25, §5, Problem 2] asks: "Find any criteria for the property (P) for finite systems of orthogonal polynomials, like the Krawtchouk polynomials." He then

introduces a symmetric system Q_n built from the Krawtchouk recurrence, notes that it has property (P) for $p_0 \le 1/2$, and asks for a general criterion that would handle this case.

The main results answer this for arrays whose ratio of downward coefficients to the profile is constant (Theorem A) or increases with the index (Theorem B). For a normalized recurrence (2.2), whose coefficients have row sums one, write $d_i = b_i + c_i$ for the profile, $\rho(i) = c_i/d_i$ for the ratio and $\nu_k = \Delta^k d(0) = \sum_{i\le k}(-1)^{k-i}\binom{k}{i}d_i$ for the Newton coefficients of the profile.

Theorem A *(Theorem 3.1).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are nonnegative and log-concave without internal zeros, let $0 < r \le 1$, and put $a_i = 1 - d_i$, $b_i = (1-r)d_i$ and $c_i = rd_i$. Then every product R_mR_n, including those with $m + n > N$, is a nonnegative combination of $R_0, \dots, R_N$ on the orthogonality support.*

The input is the profile alone: one computes the N numbers ν_k and checks their signs and $N-2$ quadratic inequalities, without forming any product coefficient. For instance, $d = \frac{1}{97}(0, 8, 24, 52, 97)$ has $\nu = \frac{1}{97}(8, 8, 4, 1)$, which is log-concave, so all its constant-ratio arrays have property (P); this profile is neither Krawtchouk, nor geometric, nor quadratic, and Section 3 embeds it in a one-parameter class of such profiles.

Theorem B *(Theorem 7.5).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k\binom{x}{k}$, where $(\rho_k)_{k\ge0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization, including the products beyond the terminal degree.*

For constant ρ, Theorem B is Theorem A. On the linear profile $d_i = i/N$, a linear ratio $\rho(i) = r + \kappa i/N$ with $\kappa \ge 0$ has Newton coefficients $(r, \kappa/N)$, so Theorem B applies whenever b is nondecreasing, that is, whenever $\kappa \le \frac{N}{2N-1}(1-r)$. On every admissible profile, the exponential ratios $\rho(x) = r(1+\varepsilon)^x$, whose Newton coefficients $r\varepsilon^k$ are geometric, are admitted whenever b is nondecreasing. Separate results treat cases that Theorem B does not cover. Ratios whose Newton coefficients are not log-concave are admitted when they are convex quadratic, including those with $t^2 < r\tau$ (Theorem 6.2), or cubic with $r\sigma \le \frac{7}{4}t\tau$ (Theorem 7.6 and Corollary 7.7). Profiles whose Newton coefficients are not log-concave are admitted when the Newton coefficients of c are log-concave and those of b are those of c times nonincreasing weights (Theorem 5.1 and Corollary 5.2). For arrays quadratic in the index (Proposition 5.3) and for profiles of Newton degree at most two (Theorem 7.1), weaker conditions suffice. Section 8 summarizes the hierarchy in two tables. Nondecreasing b alone, or nonnegative Newton coefficients of the ratio alone, do not suffice (Appendix C).

Property (P) of the classical examples is known; what is new is a coefficient criterion that recovers it, its extension to increasing ratios, and a reduction for symmetric finite systems. Theorem A contains the full positive Krawtchouk range (Corollary 3.6), whose property (P) is classical [8, 7, 25], and the geometric arrays of Corollary 3.7, the dual q-Krawtchouk systems of Koelink's construction [11]. For the second part of Szwarc's problem, a quadratic decomposition reduces property (P) of a symmetric finite system to that of

Table 1: Theorems A and B and earlier criteria for nonnegative linearization, as they apply to finite systems.

Criterion	Hypothesis	Products of a finite system
Theorem A	inequalities on the Newton coefficients of the profile; constant ratio b_i/c_i	all, including degree sum $> N$
Theorem B	inequalities on the Newton coefficients of the profile and of the ratio c_i/d_i; b nondecreasing	all, including degree sum $> N$
Askey [1, Part II]	conditions on the recurrence coefficients, through a maximum principle	degree sum $\leq N$
Szwarc [25, Thms. 3.2, 3.3]	monotonicity conditions for infinite systems	not applicable: some conditions fail at the terminal index
Szwarc [24, Thm. 1]	an auxiliary discrete hyperbolic problem; necessary and sufficient	all; not an inequality test on the coefficients
Młotkowski–Szwarc [16]	chain-sequence conditions for infinite systems on convergent point sequences	not applicable (infinite systems)
Miclo [14, Prop. 3.3]	spectral; necessary and sufficient	all; not an inequality test on the coefficients
Miclo [15, Thm. 31]	inequality conditions for chains symmetric under $i \mapsto N - i$	all; excludes the Krawtchouk systems with $N > 2$

two nonsymmetric ones (Theorem 4.1); with Theorem A it gives a parity-split criterion (Corollary 4.2) that covers Q_n in its sharp range $0 < p_0 \leq 1/2$ (Theorem 4.3), whose positivity Szwarc states. Table 1 compares Theorems A and B with earlier criteria.

Related work. Szwarc's monotonicity criteria fail for finite systems near the terminal index, whose upward coefficient vanishes (Section 4). Miclo, having found numerically that two natural discrete versions of the log-concavity theorem of Achour and Trimèche fail, concluded that "the right notion of log-concavity of a measure (or rather of a Markov kernel) has yet to be found in the discrete setting" [14, p. 433]; his condition in [15, Theorem 31] that $2^i a_i$ be nonincreasing on $[0, N/2]$ fails for the symmetric Krawtchouk system once $N > 2$, and the systems with $p_0 \neq 1/2$ are not symmetric. For the finite systems of the Askey scheme a related question is open: Terwilliger [26, Problem 36.4] asks which real Leonard systems have only nonnegative intersection numbers. Blau and Hein [2, Theorem 1.1] determine the recurrences of the homogeneous monotonic P-polynomial table algebras, which are finite systems with nonnegative linearization in which all nontrivial elements have the same Haar weight and the recurrence coefficients are monotone.

Section 9 marks the scope of the criterion: for fixed $r < 1$, coefficient balls around $K_{N,r}$ in which every array has property (P) have radius at most $3(1 - r^2)^N$ for large N (Appendix A), so a criterion valid for all N must place these arrays near its boundary, and

Theorem A places them on its edge. For fixed N and $r \to 1$ the order is $(1-r)^N$ [17, Corollary 9.11]. Positivity of the Newton coefficients does not suffice, but log-concavity is not necessary; we pose the exact boundary as a problem.

Organization. Section 2 sets up finite normalized recurrences. Section 3 proves Theorem A through a kernel recurrence whose negative moves are dominated by two-step detours, and Section 4 treats symmetric systems and Szwarc's system Q_n. Section 5 extends the kernel recurrence to independent diagonal and downward coefficients. Section 6 resums the moves of negative weight for linear and convex quadratic ratios, under the condition that b be nondecreasing, which also appears in the criteria of Askey and Szwarc for infinite systems [25, Theorems 3.1 and 3.2] and in the finite criteria of [17]. Section 7 treats ratios of higher degree, including Theorem B, and Section 8 summarizes the criteria. Section 9 discusses the limits of the criteria and Section 10 states open problems. Appendices A and B contain the stability-radius bound and the examples for Section 9, Appendix C the counterexamples and sampled tests for the increasing ratios, and Appendix D a proof of Walkup's product theorem in the Newton basis.

Known results used, companion paper and formalization. Previously known are Proposition 7.3, which is Walkup's theorem [28] and is given a self-contained proof in the Newton basis, and the Abel argument of Lemma 7.4, which is that of Wang and Yeh [29]. The written proofs do not use the companion paper [17], which is cited for comparison only; it is complementary: it gives explicit perturbation neighborhoods, inequality and reversal criteria, and constraints at the symmetric Krawtchouk endpoint. The proofs of Theorem B and of the cubic criteria are computer-assisted. Before the proof of Theorem B, a certificate map states which steps are symbolic certificates and which are finite checks and names the Lean declarations that prove them; the supplement [21] gives the command that reruns each certificate. All theorems, corollaries and propositions are formalized in Lean 4; the section on code and formalization states which Lean proofs take routes different from the written ones.

2. Setting

For integers $j \geq 0$ put $[j]_q = 1 + q + \cdots + q^{j-1}$, with $[0]_q = 0$; in particular $[j]_1 = j$. Consider real coefficients indexed by $0 \leq i \leq N$, where $N \geq 1$, satisfying

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0). \end{gathered} \tag{2.1}$$

Set $R_{-1} = 0$, $R_0 = 1$, and determine $R_1, \ldots, R_N$ by

$$xR_i = a_i R_{i+1} + b_i R_i + c_i R_{i-1} \qquad (0 \leq i < N). \tag{2.2}$$

At $i = 0$, (2.2) gives $R_1 = x$; induction at $x = 1$, using the row sums and $a_i > 0$ for $i < N$, gives $R_i(1) = 1$ for $0 \leq i \leq N$. The terminal relation is $(x - b_N)R_N - c_N R_{N-1} = 0$ on the orthogonality support. Linearization means equality of functions on this support, rather than an unrestricted polynomial identity when the degree sum exceeds N.

Write the tridiagonal recurrence matrix of such an array with the row convention

$$J_{i,i+1} = a_i, \qquad J_{i,i} = b_i, \qquad J_{i,i-1} = c_i,$$

with all other entries zero and indices outside $0, \dots, N$ omitted. We call an array satisfying (2.1) *normalized*. Its matrix J has row sums one, but (2.1) allows $b_i < 0$; we call the array *stochastic* only when all $b_i \geq 0$, so that J is a stochastic matrix. The arrays of Theorem 3.1 below are stochastic.

Lemma 2.1 *(Finite recurrence realization and triangle support).* *Every recurrence satisfying (2.1) has a positive orthogonality measure on exactly $N+1$ real points. If J is its tridiagonal recurrence matrix, then its finite product coefficients satisfy*

$$g(m, n; k) = [R_m(J)]_{n,k}. \tag{2.3}$$

They vanish unless $|m-n| \leq k \leq \min(m+n, N)$, regardless of their signs inside this interval.

Proof. Choose positive π_i with $\pi_0 = 1$ and $\pi_{i+1} = \pi_i a_i / c_{i+1}$. For $D = \operatorname{diag}(\sqrt{\pi_i})$, $H = DJD^{-1}$ is real symmetric. Induction in (2.2) gives the row-vector identity $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$, hence $e_0^{\mathsf{T}} R_i(H) = \pi_i^{-1/2} e_i^{\mathsf{T}}$. In particular e_0 is cyclic for H. In an orthogonal eigenspace decomposition of H, the cyclic span of e_0 has dimension at most the number of distinct eigenvalues; here it is all of $\mathbb{R}^{N+1}$, so H has $N+1$ distinct eigenvalues and the projection of e_0 on each eigenspace is nonzero. The spectral measure μ of H at e_0 therefore has positive mass at each of $N+1$ real points, and

$$\int R_i R_j \, d\mu = e_0^{\mathsf{T}} R_i(H) R_j(H) e_0 = \mathbf{1}_{i=j} \pi_i^{-1}.$$

The terminal relation holds on this support: its polynomial $Q(x) = (x - b_N) R_N - c_N R_{N-1}$ satisfies $e_0^{\mathsf{T}} Q(J) = 0$. Since e_0 is cyclic and $Q(J)$ commutes with J, $Q(J) = 0$. Multiplying a basis polynomial by x acts by the corresponding row of J, including the terminal row. Induction on powers of x therefore gives (2.3). Orthogonality also gives

$$g(m, n; k) = \frac{\int R_m R_n R_k \, d\mu}{\int R_k^2 \, d\mu}.$$

If one of m, n, k exceeds the sum of the other two, its polynomial is orthogonal to their product, whose degree is smaller. This proves the asserted support. ∎

We say that the array has *nonnegative linearization*, or *property (P)*, if $g(m, n; k) \geq 0$ for all $0 \leq m, n, k \leq N$. The coefficients $g(m, n; \cdot)$ always sum to one, since $R_i(1) = 1$ and 1 lies in the support, because $J\mathbf{1} = \mathbf{1}$.

For $q \geq 1$ and $0 < r \leq 1$ let $G_{N,q,r}$ be the array

$$a_i = \frac{q^i [N-i]_q}{[N]_q}, \qquad b_i = (1-r)\frac{[i]_q}{[N]_q}, \qquad c_i = r\frac{[i]_q}{[N]_q}, \tag{2.4}$$

with $[j]_q$ as above; it satisfies (2.1), since $[N]_q - [i]_q = q^i [N-i]_q$. Its profile $d_i = [i]_q / [N]_q$ has geometric increments with ratio q, and $b_i / c_i = (1-r)/r$ is constant. The Krawtchouk array is $K_{N,r} = G_{N,1,r}$, with $a_i = (N-i)/N$, $b_i = (1-r)i/N$ and $c_i = ri/N$.

3. Log-concave profiles

For an array satisfying (2.1) with profile $d_i = b_i + c_i$ write

$$\nu_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} d_i \quad (1 \le k \le N), \qquad d_n = \sum_{k=1}^{n} \binom{n}{k} \nu_k \quad (0 \le n \le N);$$

below we put $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$. We say that $(\nu_k)_{1 \le k \le N}$ is *log-concave without internal zeros* if all $\nu_k \ge 0$, the indices with $\nu_k > 0$ form an interval $\{1, \dots, K\}$, and $\nu_k^2 \ge \nu_{k-1}\nu_{k+1}$ for $2 \le k \le N-1$. Note $\nu_1 = d_1 > 0$.

Theorem 3.1 *(Log-concave Newton criterion). Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$, and put $a_i = 1 - d_i$, $b_i = (1-r)d_i$, $c_i = rd_i$. Then the system has nonnegative linearization.*

The hypotheses are inequalities in the profile. They hold for the Krawtchouk arrays ($\nu = (1/N, 0, \dots, 0)$) and for all arrays $G_{N,q,r}$ (ν_k proportional to $(q-1)^{k-1}$, log-concave with equality). The hypotheses also hold for every admissible convex quadratic profile: $\nu_1 > 0$, $\nu_2 \ge 0$ when $N \ge 2$, and $\nu_k = 0$ for $3 \le k \le N$. The hypotheses exclude the known counterexamples with constant ratio: the convex counterexamples recorded in [17] have Newton coefficients of both signs, and for the combinations $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ of Section 9, with $q > 1$, $N \ge 3$ and $0 < \lambda < 1$, one has $\nu_1 = (1-\lambda)/N + \lambda/[N]_q$ and $\nu_k = \lambda(q-1)^{k-1}/[N]_q$ for $2 \le k \le N$, so $\nu_2^2 - \nu_1\nu_3 = -\lambda(1-\lambda)(q-1)^2/(N[N]_q) < 0$.

Applying Theorem 3.1. Given a profile $0 = d_0 < d_1 < \cdots < d_N = 1$:

1. compute $\nu_k = \sum_{i \le k} (-1)^{k-i} \binom{k}{i} d_i$ for $1 \le k \le N$;
2. check that all $\nu_k \ge 0$, that the k with $\nu_k > 0$ form an interval $\{1, \dots, K\}$, and that $\nu_k^2 \ge \nu_{k-1}\nu_{k+1}$ for $2 \le k \le N-1$;
3. conclude property (P), for all products $R_m R_n$ with $0 \le m, n \le N$, for $a_i = 1 - d_i$, $b_i = (1-r)d_i$, $c_i = rd_i$ and every $0 < r \le 1$.

For $d = \frac{1}{97}(0, 8, 24, 52, 97)$: $97\nu_1 = 8$, $97\nu_2 = 24 - 16 = 8$, $97\nu_3 = 52 - 72 + 24 = 4$, $97\nu_4 = 97 - 208 + 144 - 32 = 1$. All are positive, and $8^2 \ge 8 \cdot 4$, $4^2 \ge 8 \cdot 1$. So every constant-ratio array with this profile has property (P); at $r = 1$, for instance, $g(4, 4; 4) = 95753/292365$.

This profile is a member of a one-parameter class. For $\alpha > 0$ and $0 < q \le 1$ let $\nu_k = C\alpha^{k-1} q^{\binom{k-1}{2}}$ for $1 \le k \le N$, with $C > 0$ chosen so that $d_N = \sum_k \binom{N}{k} \nu_k = 1$. Then $\nu_k^2 = q^{-1}\nu_{k-1}\nu_{k+1} \ge \nu_{k-1}\nu_{k+1}$, so Theorem 3.1 applies. For $q = 1$ these are the profiles of the geometric arrays $G_{N,1+\alpha,r}$, and as $q \to 0$ they tend to quadratic profiles. For $0 < q < 1$ and $N \ge 3$ the increments are not geometric and $\nu_3 \ne 0$, so the arrays are neither Krawtchouk, nor geometric, nor quadratic. The profile in the box is $N = 4$, $\alpha = 1$, $q = \frac{1}{2}$.

Outline of the proof. The proof has four steps. (1) A rescaling removes r from the downward coefficients, and in the basis $E_i = e_i p_i$ the product coefficients become positive

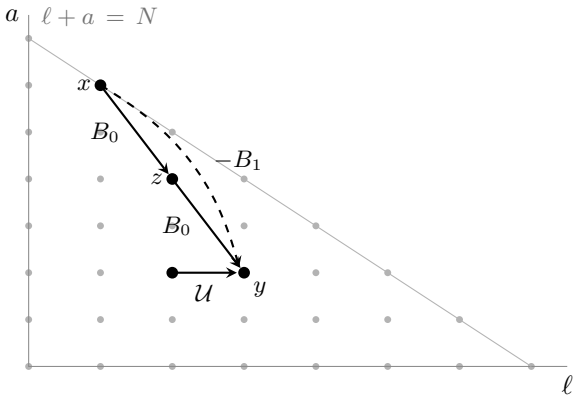


Figure 1: The kernel recurrence of Lemma 3.2 on the triangle Ω ($N = 7$). Every move raises ℓ. The node $y = (\ell + 1, a)$ receives the straight move $\mathcal{U}$ from (ℓ, a) and moves of index j from nodes with smaller ℓ; the moves of odd index carry a minus sign. The dashed move $-B_1$ from x, of displacement $(2, -4)$, is dominated by a quarter of the two-step paths from x through $z = x + (1, -2)$ to y, made of two moves B_0 of even index (Lemma 3.4); this gives $\mathcal{P}^2 \geq 4\mathcal{Q}$.

Table 2: The negative moves into $y = (\ell + 1, a)$, for odd j, and the two-step paths through moves of even index that dominate them (Lemma 3.4). The first step of a detour has index 0, the second has index $j - 1$.

Negative move	Source node	Dominating detours
tA_j	$(\ell - j,\ a + 2j + 1)$	A_0 then B_{j-1}; B_0 then A_{j-1}
B_j	$(\ell - j,\ a + 2j + 2)$	B_0 then B_{j-1}
C_j	$(\ell - j - 1,\ a + 2j + 2)$	B_0 then C_{j-1}

multiples of numbers L^k_{mn}. For fixed k these form a symmetric generating function $K_k(z, w)$, which an explicit difference operator annihilates; the operator includes the terminal row, so the argument also covers products of degree sum greater than N. Expanding $K_k = \sum \omega_{\ell,a}(z + w)^a (zw)^\ell$, it suffices to show $\omega_{\ell,a} \geq 0$. (2) Lemma 3.2 turns the annihilation into a recurrence that expresses ω at level $\ell + 1$ through lower levels, by *moves* with explicit weights $a_{\ell+a}$, tA_j, B_j and C_j; the moves of odd index j carry a minus sign. The recurrence holds for every profile, and its weights are nonnegative when all $\nu_k \geq 0$. (3) Log-concavity enters only through Lemmas 3.3 and 3.4: each move of odd index j is dominated by a quarter of the two-step paths through moves of the even indices 0 and $j - 1$. (4) In operator form, $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\mathcal{P}^2 \geq 4\mathcal{Q}$, and induction on ℓ gives $\omega \geq \frac{1}{2}\mathcal{P}\omega \geq 0$. Figure 1 shows the moves and one detour, and Table 2 lists all detours.

A rescaled system and a kernel. Fix r and put $t = (1 - r)/\sqrt{r} \geq 0$. If p_i is defined by $p_0 = 1$ and $yp_i = a_i p_{i+1} + t d_i p_i + d_i p_{i-1}$ $(0 \leq i < N)$, then $R_i(x) = r^{i/2} p_i(x/\sqrt{r})$, so the coefficients H^k_{mn} in $p_m p_n = \sum_k H^k_{mn} p_k$ on the support satisfy $g(m, n; k) =$

$r^{(m+n-k)/2}H^k_{mn}$. It suffices to show $H^k_{mn} \geq 0$. Put

$$e_i \,=\, \prod_{j=0}^{i-1} \frac{a_j}{d_{j+1}}, \qquad E_i \,=\, e_i p_i.$$

Then $yE_i \,=\, d_{i+1}E_{i+1} + td_iE_i + a_{i-1}E_{i-1}$ on the support for $0 \leq i \leq N$, with the terms $d_{N+1}E_{N+1}$ and $a_{-1}E_{-1}$ omitted; at $i = N$ this is the terminal relation. Write $E_mE_n \,=\, \sum_k L^k_{mn}E_k$, so that $H^k_{mn} \,=\, L^k_{mn}e_k/(e_me_n)$, and

$$K_k(z,w) \,=\, \sum_{m,n=0}^{N} L^k_{mn} z^m w^n.$$

Let $\theta_z \,=\, z\,\partial_z$ and let $d(\theta) \,=\, \sum_k \nu_k \binom{\theta}{k}$ act on polynomials by $d(\theta_z)z^i \,=\, d_i z^i$. For $F(z) \,=\, \sum_{i=0}^{N} E_i z^i$ the recurrence reads $yF(z) \,=\, \mathcal{L}_z F(z)$ with

$$\mathcal{L}_z \,=\, (z^{-1} + t - z)\, d(\theta_z) + z,$$

because $d_0 \,=\, 0$ removes the term $z^{-1}E_0$ and $1 - d_N \,=\, 0$ removes the term $z^{N+1}E_N$. Applying this to $F(z)F(w) \,=\, \sum_k K_k(z,w)E_k$ in the two variables separately and comparing coefficients of the basis E_k gives $(\mathcal{L}_z - \mathcal{L}_w)K_k \,=\, 0$. Since K_k is symmetric, $K_k \,=\, \sum_{\ell,a\geq 0} \omega_{\ell,a} S^a Z^\ell$ with $S \,=\, z + w$, $Z \,=\, zw$, and $K_k(z,0) \,=\, z^k$ gives $\omega_{0,a} \,=\, \mathbf{1}_{a=k}$. As K_k has degree at most N in z, comparing the coefficients of $z^{\ell+a}w^\ell$ by induction on ℓ gives $\omega_{\ell,a} \,=\, 0$ whenever $\ell + a \,>\, N$.

The kernel recurrence. Put $f_n(\ell) \,=\, \sum_{h\geq 0} \binom{\ell}{h} \nu_{n+h}$ with $\nu_0 \,=\, 0$ and $\nu_k \,=\, 0$ for $k \,>\, N$, so $f_0(\ell) \,=\, d_\ell$, and

$$W_{p,q}(\ell,a) \,=\, \sum_{n\geq p} \binom{n-p+q}{q} \binom{a}{n} f_n(\ell) \,\geq\, 0,$$

and $A_j \,=\, W_{2j+1,j}$, $B_j \,=\, W_{2j+2,j}$, $C_j \,=\, W_{2j+2,j+1}$.

Lemma 3.2. *For all $\ell, a \,\geq\, 0$,*

$$\begin{aligned} d_{\ell+1}\omega_{\ell+1,a} \,=\, a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\geq 0} (-1)^j \Big(& t\,A_j(\ell-j, a+2j+1)\,\omega_{\ell-j,a+2j+1} \\ & + B_j(\ell-j, a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ & + C_j(\ell-j-1, a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned} \tag{3.1}$$

where $a_i \,=\, 1 - d_i$ is extended by the polynomial d_i for $i \,>\, N$ and terms with a negative index are omitted.

Proof. Since $S^aZ^\ell \,=\, \sum_i \binom{a}{i} z^{\ell+i} w^{\ell+a-i}$, Vandermonde's identity $\binom{\ell+i}{k} \,=\, \sum_{p+q=k} \binom{\ell}{q}\binom{i}{p}$ and $\binom{a}{i}\binom{i}{p} \,=\, \binom{a}{p}\binom{a-p}{i-p}$ give

$$d(\theta_z)\, S^a Z^\ell \,=\, \sum_{p,q\geq 0} \nu_{p+q} \binom{\ell}{q} \binom{a}{p} z^p S^{a-p} Z^\ell.$$

Hence $(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell$ equals $(z-w)S^aZ^\ell$ plus $\sum_{p,q}\nu_{p+q}\binom{\ell}{q}\binom{a}{p}S^{a-p}Z^\ell$ times $(z^{p-1} - w^{p-1}) + t(z^p - w^p) - (z^{p+1} - w^{p+1})$. Divide by $z-w$, using $\chi_n = (z^{n+1} - w^{n+1})/(z - w) = \sum_{0\le j\le\lfloor n/2\rfloor}(-1)^j\binom{n-j}{j}S^{n-2j}Z^j$ for $n \ge 0$, $\chi_{-1} = 0$ and $(z^{-1} - w^{-1})/(z-w) = -Z^{-1}$. Since $\sum_q \binom{\ell}{q}\nu_{p+q} = f_p(\ell)$, the quotient is

$$
\begin{aligned}
&- d_\ell S^aZ^{\ell-1} + a_{\ell+a}S^aZ^\ell \\
&+ \sum_{j\ge0}(-1)^j\Big(tA_j(\ell,a)S^{a-2j-1}Z^{\ell+j} + B_j(\ell,a)S^{a-2j-2}Z^{\ell+j} + C_j(\ell,a)S^{a-2j-2}Z^{\ell+j+1}\Big).
\end{aligned}
$$

Here $-d_\ell S^aZ^{\ell-1}$ comes from $p=0$; the term $a_{\ell+a}S^aZ^\ell$ combines S^aZ^ℓ with the parts $j=0$ of $-\chi_p$, which sum to $-d_{\ell+a}S^aZ^\ell$ by Vandermonde; and the three families from χ_{p-2}, $t\chi_{p-1}$ and $-(\chi_p - S^p)$, whose coefficients $\binom{p-2-j}{j}$, $\binom{p-1-j}{j}$ and $\binom{p-1-j}{j+1}$ are those of B_j, A_j and C_j. Summing over ℓ, a with weights $\omega_{\ell,a}$ and taking the coefficient of S^aZ^ℓ gives (3.1), because the monomials S^aZ^ℓ are linearly independent. ∎

The log-concavity inequality.

Lemma 3.3. *Let $(u_k)_{k\ge0}$ be nonnegative, finitely supported and log-concave without internal zeros, that is, its positive entries form an interval and $u_k^2 \ge u_{k-1}u_{k+1}$ for all $k \ge 1$, and put $f_n^u(\ell) = \sum_{h\ge0}\binom{\ell}{h}u_{n+h}$. Then $(u_k + u_{k+1})_{k\ge0}$ has the same property, hence so does $(f_n^u(\ell))_{n\ge0}$ for every $\ell \ge 0$, and*

$$f_i^u(\ell)\, f_l^u(\ell+1) \;\ge\; f_0^u(\ell+1)\, f_{i+l}^u(\ell) \qquad (i,l \ge 1,\ \ell \ge 0).$$

In particular, under the hypotheses of Theorem 3.1, $f_i(\ell)\, f_l(\ell+1) \ge d_{\ell+1}\, f_{i+l}(\ell)$ for $i,l \ge 1$ and $\ell \ge 0$.

Proof. For a nonnegative finitely supported sequence u that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_iu_{i+2}) + (u_iu_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(f_n^u(\ell))_n$ is obtained from $(u_n)_n$ by applying $u \mapsto v$ ℓ times. For such a sequence g, balanced products dominate: $g_ig_l \ge g_0g_{i+l}$ and $g_ig_{l+1} \ge g_1g_{i+l}$ for $i,l \ge 1$. Adding these with $g = f^u(\ell)$ and using $g_l + g_{l+1} = f_l^u(\ell+1)$ and $g_0 + g_1 = f_0^u(\ell+1)$ gives the claim. Under the hypotheses of Theorem 3.1 the sequence $u = (0, \nu_1, \nu_2, \ldots)$ has the property, $f^u = f$, and $f_0(\ell+1) = d_{\ell+1}$. ∎

Lemma 3.4. *Under the hypotheses of Theorem 3.1, for all $j \ge 1$ and $\ell, a \ge 0$,*

$$
\begin{aligned}
A_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}A_j(\ell,a),\\
B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}B_j(\ell,a),\\
B_0(\ell,a)C_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}C_j(\ell,a),
\end{aligned}
$$

with $W_{p,q}(\ell,a) = 0$ for $a < 0$.

Proof. Use $A_0(\ell,a) \ge af_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}f_2(\ell)$, expand the second factor, apply Lemma 3.3 to $f_1(\ell)f_n(\ell+1)$ and $f_2(\ell)f_n(\ell+1)$, and use $a\binom{a-1}{n} = (n+1)\binom{a}{n+1}$ and

$\binom{a}{2}\binom{a-2}{n} = \binom{n+2}{2}\binom{a}{n+2}$. After division by $d_{\ell+1}$ the three left sides are bounded below by $\sum_m \gamma_m \binom{a}{m} f_m(\ell)$ with

$$\gamma_m = \binom{m+1}{2}\binom{m-j-2}{j-1}, \qquad \binom{m}{2}\binom{m-j-3}{j-1}, \qquad \binom{m}{2}\binom{m-j-2}{j},$$

while the right sides are $4\sum_m \gamma'_m \binom{a}{m} f_m(\ell)$ with $\gamma'_m = \binom{m-j-1}{j}$, $\binom{m-j-2}{j}$ and $\binom{m-j-1}{j+1}$. Where $\gamma'_m > 0$ the quotients are

$$\frac{jm(m+1)}{2(m-j-1)}, \qquad \frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)},$$

on $m \geq 2j+1$, $m \geq 2j+2$ and $m \geq 2j+2$ respectively. They increase with j for fixed m; at $j = 1$ their minima over integers are 5, 5 and 6. All quotients exceed 4, and $f_m(\ell) \geq 0$. ∎

Proof of Theorem 3.1. Let $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, on which ω is supported. Read (3.1) as moves into the node $(\ell+1, a)$, each divided by $d_{\ell+1} > 0$: the straight move from (ℓ, a) with weight $a_{\ell+a} \geq 0$ on Ω, and for each j the three moves from $(\ell - j, a + 2j + 1)$, $(\ell - j, a + 2j + 2)$, $(\ell - j - 1, a + 2j + 2)$ with weights tA_j, B_j, C_j. Every move raises ℓ. Let $\mathcal{U}$ be the straight moves, $\mathcal{P}$ the moves with j even and $\mathcal{Q}$ the moves with j odd, all with nonnegative weights, so that $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\beta = \omega|_{\ell=0}$.

We claim $\mathcal{P}^2 \geq 4\mathcal{Q}$ entrywise. A move of $\mathcal{Q}$ with index j from $x = (\ell, a)$ is matched by the two-step paths in $\mathcal{P}^2$ that start with A_0 or B_0 from x (index 0, landing at $(\ell+1, a-1)$ or $(\ell+1, a-2)$, which lie in Ω) and continue with the move of the same type and index $j-1$ (even), as in Lemma 3.4: A_j by A_0 then B_{j-1} and B_0 then A_{j-1}; B_j by B_0 then B_{j-1}; C_j by B_0 then C_{j-1}. These paths end at the target of the matched move, carry the same power of t, and have the extra denominator $d_{\ell+1}$ of the intermediate node. Distinct moves of $\mathcal{Q}$ have distinct displacements, so their paths are distinct, and Lemma 3.4 gives the claim.

Now show by induction on the level that $\omega \geq 0$ and $\omega \geq \frac{1}{2}\mathcal{P}\omega$. Both hold at $\ell = 0$. If they hold below a node y at level $\ell + 1 \geq 1$, then, all weights being nonnegative, $(\mathcal{P}\omega)(y) \geq \frac{1}{2}(\mathcal{P}^2\omega)(y) \geq 2(\mathcal{Q}\omega)(y)$, and

$$\omega(y) = (\mathcal{U}\omega)(y) + (\mathcal{P}\omega)(y) - (\mathcal{Q}\omega)(y) \geq \tfrac{1}{2}(\mathcal{P}\omega)(y) \geq 0.$$

Hence all $\omega_{\ell,a} \geq 0$, so K_k has nonnegative coefficients in z, w, so $L^k_{mn} \geq 0$, $H^k_{mn} \geq 0$ and $g(m, n; k) \geq 0$. ∎

Remark 3.5. The argument is valid coefficientwise in t. With independent parameters in $yp_i = a_i p_{i+1} + ud_i p_i + vd_i p_{i-1}$, weighted homogeneity ($\deg y = \deg u = 1$, $\deg v = 2$) then shows that each linearization coefficient is a polynomial in (u, v) with nonnegative coefficients: for $D = m + n - k \geq 0$ it has the form $\sum_{0 \leq j \leq \lfloor D/2 \rfloor} h_j u^{D-2j} v^j$ with $h_j \geq 0$, and it vanishes for $D < 0$. Consequently every $g(m, n; k)$ of the array of Theorem 3.1, as a function of r, equals $\sum_j h_j r^j (1-r)^{D-2j}$; multiplying the j-th term by $(r + (1-r))^j = 1$ shows that its coefficients in the Bernstein basis of degree D are $\binom{D}{s}^{-1} \sum_{j \leq s \leq 2j} h_j \binom{j}{s-j} \geq 0$.

Szwarc [25, §5, Problem 2] records the positive range of the Krawtchouk polynomials as $0 < p_0 \le 1/2$.[1]

Corollary 3.6 *(Full positive Krawtchouk range).* *For every $N \ge 1$ and $0 < p_0 \le 1/2$, Krawtchouk polynomials orthogonal for binomial weights $\binom{N}{j} p_0^{N-j}(1-p_0)^j$ have nonnegative linearization when normalized at $j = 0$.*

Proof. To fix the convention without relying on a named recurrence, put $r = p_0/(1-p_0) \in (0,1]$, and let $Y_1, \dots, Y_N$ be independent variables with values $1, -r$ of probabilities $p_0, 1-p_0$. Their mean is zero and $Y_i^2 = (1-r)Y_i + r$. The normalized elementary symmetric sums

$$K_i(j) = \binom{N}{i}^{-1} \sum_{|S|=i} \prod_{l \in S} Y_l$$

depend only on the number j of negative coordinates. They are orthogonal, because the expectation of two distinct squarefree monomials is zero: some independent mean-zero variable occurs once. Their weights in j are exactly the stated binomial weights. With $x = N^{-1}\sum_l Y_l = 1 - j/(N(1-p_0))$, multiplication by x gives

$$xK_i = \frac{N-i}{N} K_{i+1} + (1-r)\frac{i}{N} K_i + r\frac{i}{N} K_{i-1}.$$

Indeed, in multiplying the elementary sum by $\sum_l Y_l$, indices outside a subset give $(i+1)$ copies of each subset of size $i+1$; the linear part of Y_l^2 gives $(1-r)i$ copies of each subset of size i; and the constant part gives $r(N-i+1)$ copies of each subset of size $i-1$. Division by the binomial normalizers gives the displayed recurrence, including $i = N$. Its leading coefficient in x is positive, its degree is i, and its value at $j = 0$ is one. These properties identify the normalized orthogonal polynomials for the binomial measure. This is the array $K_{N,r}$; its profile $d_i = i/N$ has Newton coefficients $(1/N, 0, \dots, 0)$, so Theorem 3.1 applies. The range is sharp in this endpoint normalization: if $1/2 < p_0 < 1$, then $r = p_0/(1-p_0) > 1$ and $g(1,1;1) = b_1 = (1-r)/N < 0$. For $N = 1$, this coefficient follows from the terminal relation. ∎

Corollary 3.7 *(Geometric arrays).* *For every $N \ge 2$, $q > 1$, and $0 < r \le 1$, the array $G_{N,q,r}$ of (2.4) has nonnegative linearization. In the normalization $R_0 = 1$, $R_1 = x$, it is outside the Krawtchouk recurrence class.*

Proof. Its increments are $d_i - d_{i-1} = q^{i-1}/[N]_q$, so its Newton coefficients $\nu_k = (q-1)^{k-1}/[N]_q$ are log-concave without internal zeros and Theorem 3.1 applies. For a Krawtchouk recurrence in this normalization the increments of $b_i + c_i$ are constant. Here their ratio is $q > 1$ and $N \ge 2$, so they are not constant. An affine change of the support transporting the same endpoint-normalized basis leaves the multiplication coefficients in the basis unchanged; it cannot remove this distinction. ∎

[1] In the author version of [25], §5, Problem 2, printed p. 125, the displayed weights $\binom{N}{n} p^{N-n}(1-p)^n$ have mean $N(1-p)$, whereas the displayed recurrence gives $K_1(x) = (x - pN)/((1-p)N)$, so that $\int K_1 \, d\mu = (1-2p)/(1-p)$. The corollary fixes the convention directly: it uses the displayed weights and the normalization at the point carrying the weight p^N.

The arrays $G_{N,q,r}$ are the normalized recurrences of Koelink's construction [11, Theorems 3.1(ii) and 7.1] with $r = 1/p$; their polynomial interpretation is the dual q-Krawtchouk interpretation described after his Theorem 7.8. A proof of Corollary 3.7 by a finite algebra with positive multiplication, which needs only the geometric increments, is given in [17].

For some parameters both families come from distance-regular graphs. Dividing the intersection array of such a graph by its valency gives an array satisfying (2.1), and its linearization coefficients are positive multiples of the intersection numbers p^k_{mn}, hence nonnegative [4, 6]. For an integer $s \geq 2$, the Hamming graph $H(N, s)$ gives $K_{N,r}$ with $r = 1/(s-1)$; a dual polar graph of diameter N with classical parameters $(N, q, 0, q^e)$ gives $G_{N,q,r}$ with $r = q^{-e}$. Corollaries 3.6 and 3.7 cover the full parameter ranges.

4. Szwarc's quadratic system

Problem 2 of [25, §5] does not stop at the Krawtchouk recurrence. With his parameter written as p_0, Szwarc introduces the symmetric recurrence of length $2N+1$

$$yQ_{2n} = (N-n)Q_{2n+1} + nQ_{2n-1}, \qquad yQ_{2n-1} = (1-p_0)Q_{2n} + p_0 Q_{2n-2}, \tag{4.1}$$

records that $Q_{2n}(y)$ is the Krawtchouk polynomial $K_n(y^2)$ of his recurrence, states that the Q_n have nonnegative linearization for $p_0 \leq 1/2$, and asks for a general criterion that would handle this case. His Theorem 3.3 for infinite symmetric systems does not apply. In the normalization (4.1) its monotonicity hypotheses hold, and its hypothesis $\alpha_n \leq \gamma_n$ holds at odd indices exactly when $p_0 \leq 1/2$; at an even index $2n$ it reads $n \leq N - n$ and fails in the upper half. Some failure of this kind is forced in every finite system, whose terminal upward coefficient vanishes. We derive the positivity of (4.1) from Theorem 3.1 through a quadratic decomposition of symmetric finite systems.

Put $M = 2N$, $R_{2n}(t) = Q_{2n}(\sqrt{N}t)$ and $R_{2n-1}(t) = \sqrt{N}\,Q_{2n-1}(\sqrt{N}t)$. Substitution in (4.1) shows that the R_k satisfy (2.2), with N replaced by M, for the array

$$b_k = 0, \qquad (a_{2n}, c_{2n}) = \Big(\frac{N-n}{N}, \frac{n}{N}\Big), \qquad (a_{2n-1}, c_{2n-1}) = (1-p_0,\, p_0). \tag{4.2}$$

It satisfies (2.1) with M in place of N whenever $0 < p_0 < 1$. The two normalizations differ by positive factors and a positive dilation, so they have the same product coefficients up to positive factors.

Szwarc uses the quadratic transformation [25, §3.5] to pass from a symmetric system to its even part. In [22] a parity-split monotonicity criterion for symmetric infinite systems gives property (P) of the generalized Chebyshev polynomials and, through their even parts, Gasper's result for the Jacobi polynomials with $\alpha \geq \beta$ and $\alpha + \beta + 1 \geq 0$; in [23] the same route gives property (P) of Askey–Wilson polynomials for a range of parameters. In both, positivity passes from a symmetric system to its even part. For the finite systems of classical association schemes, bipartite halves and antipodal quotients give quadratic transformations of the polynomials, which Chihara and Stanton [5] derive from a quadratic transformation of the Askey–Wilson polynomials. The decomposition of a symmetric system into its even and odd components, and the identities relating their products, are standard, and we claim

no priority for them. What we use is the converse implication for finite systems: positivity of a symmetric system, including the products beyond the terminal degree, follows from positivity of its two components, both read off from its coefficients, which need not come from a scheme. Any criterion for nonsymmetric finite systems therefore becomes a criterion for symmetric ones. For a family of infinite systems a result of this kind is known: Kahler [10] solved Problem 1 of [25, §5] by showing that the generalized Chebyshev polynomials, the quadratic transforms of the Jacobi polynomials, have property (P) exactly for the parameters for which the Jacobi polynomials $R_n^{(\alpha,\beta)}$ have it. His result concerns that family; Theorem 4.1 below is a reduction for arbitrary finite symmetric systems.

Theorem 4.1 *(Quadratic decomposition).* *Let $N \geq 2$, and let a_k, c_k $(0 \leq k \leq 2N)$ with $b \equiv 0$ satisfy (2.1) with $2N$ in place of N. Put $a_k = c_k = 0$ for $k \notin [0, 2N]$, and define the even component*

$$\hat{a}_n = a_{2n}a_{2n+1}, \quad \hat{b}_n = c_{2n}a_{2n-1} + a_{2n}c_{2n+1}, \quad \hat{c}_n = c_{2n}c_{2n-1} \qquad (0 \leq n \leq N)$$

and the odd component

$$\check{a}_n = a_{2n+1}a_{2n+2}, \quad \check{b}_n = a_{2n+1}c_{2n+2} + c_{2n+1}a_{2n}, \quad \check{c}_n = c_{2n+1}c_{2n} \qquad (0 \leq n < N).$$

After the affine normalization $(\hat{a}_n, \hat{b}_n, \hat{c}_n) \mapsto (\hat{a}_n, \hat{b}_n - \hat{b}_0, \hat{c}_n)/\hat{a}_0$, and likewise for the odd component, these arrays satisfy (2.1) on the indices $0, \dots, N$ and $0, \dots, N-1$. If both have nonnegative linearization, so does the symmetric array. More precisely, with $\hat{g}$ and $\check{g}$ the product coefficients of the components and terms with indices out of range omitted,

$$g(2a, 2b; 2c) = \hat{g}(a, b; c), \qquad g(2a, 2b+1; 2c+1) = a_{2a}\,\check{g}(a, b; c) + c_{2a}\,\check{g}(a-1, b; c),$$

the coefficients with $m + n + k$ odd vanish, and $g(2a+1, 2b+1; 2c) \int R_{2c}^2\, d\mu = g(2c, 2a+1; 2b+1) \int R_{2b+1}^2\, d\mu$. Conversely, nonnegative linearization of the symmetric array implies that of its even component.

Proof. Since $b \equiv 0$, the matrix J is conjugate to $-J$ by $\operatorname{diag}((-1)^i)$, which fixes e_0; hence the measure of Lemma 2.1 is symmetric and $R_k(-t) = (-1)^k R_k(t)$. By symmetry $R_{2n}(t) = P_n(t^2)$ and $R_{2n+1}(t) = tL_n(t^2)$ for polynomials P_n, L_n of degree n. Applying (2.2) twice gives, with $x = t^2$,

$$xP_n = \hat{a}_n P_{n+1} + \hat{b}_n P_n + \hat{c}_n P_{n-1}, \qquad xL_n = \check{a}_n L_{n+1} + \check{b}_n L_n + \check{c}_n L_{n-1}.$$

The row sums of the components equal $a_{2n} + c_{2n} = 1$ and $a_{2n+1} + c_{2n+1} = 1$; the coefficients $\hat{a}_n, \check{a}_n$ are positive below the last index and vanish there, and $\hat{c}_n, \check{c}_n > 0$ for $n \geq 1$. Hence the normalized arrays, which describe the same polynomials in the variable $(x - \hat{b}_0)/\hat{a}_0$ (respectively $(x - \check{b}_0)/\check{a}_0$), satisfy (2.1). The terminal relation $tR_{2N} = c_{2N}R_{2N-1}$ on the support Σ of the symmetric system, applied once and twice, gives the terminal relations of P on $\{t^2 : t \in \Sigma\}$ and of L on $\{t^2 : t \in \Sigma,\ t \neq 0\}$. These sets have $N+1$ and N points, because Σ is symmetric with $2N+1$ points and contains 0. By Lemma 2.1 they are the supports of the two components.

The identity $tR_{2n} = a_{2n}R_{2n+1} + c_{2n}R_{2n-1}$ gives $P_n = a_{2n}L_n + c_{2n}L_{n-1}$ on the support of L; for $n = N$ it is the terminal relation, with $a_{2N} = 0$. Hence, on $\Sigma \setminus \{0\}$,

$$R_{2a}R_{2b+1} = t\big(a_{2a}L_aL_b + c_{2a}L_{a-1}L_b\big) = \sum_c \big(a_{2a}\check{g}(a,b;c) + c_{2a}\check{g}(a-1,b;c)\big)R_{2c+1},$$

and both sides vanish at $t = 0$. Similarly $R_{2a}R_{2b} = (P_aP_b)(t^2) = \sum_c \hat{g}(a,b;c)R_{2c}$ on Σ. Uniqueness of expansions on Σ gives the displayed formulas. The vanishing for $m+n+k$ odd and the relation for two odd indices follow from the symmetry of the measure and from $g(m,n;k)\int R_k^2\,d\mu = \int R_mR_nR_k\,d\mu$. The converse holds because $\hat{g}(a,b;c) = g(2a,2b;2c)$. ∎

Combined with Theorem 3.1, the decomposition gives a parity-split criterion of the same shape as Szwarc's Theorem 3.3: separate conditions on the even-indexed and the odd-indexed coefficients.

Corollary 4.2 *(Parity-split criterion).* *Let $N \geq 1$ and let a symmetric array ($b \equiv 0$) on the indices $0,\dots,2N$ satisfy (2.1). Suppose that*

1. *the odd downward coefficients are constant, $c_{2n+1} = p_0$, with $0 < p_0 \leq 1/2$;*
2. *the even downward coefficients $d_n = c_{2n}$ have positive increments $d_n - d_{n-1}$, $1 \leq n \leq N$, forming a geometric progression with ratio $q \geq 1$.*

Then the array has nonnegative linearization. With $r = p_0/(1-p_0)$ and $N \geq 2$, its normalized even and odd components are (2.4) with parameters (N,q,r) and $(N-1,q,r/q)$, respectively. For $N = 1$ the odd component consists only of $L_0 = 1$ and the array is (4.2). If $1/2 < p_0 < 1$ in (i) and (ii) holds, then $g(2,2;2) < 0$.

Proof. For $N = 1$ the array is $a_0 = 1$, $c_1 = p_0$, $a_1 = 1-p_0$, $c_2 = 1$, with support $\{-1,0,1\}$ and $R_1 = t$, $R_2 = (t^2 - p_0)/(1-p_0)$. On the support $R_1^2 = (1-p_0)R_2 + p_0$, $R_1R_2 = R_1$ and $R_2^2 = \frac{p_0}{1-p_0} + \frac{1-2p_0}{1-p_0}R_2$, as evaluation at $t = 0, \pm 1$ shows; all coefficients are nonnegative for $p_0 \leq 1/2$, and $g(2,2;2) < 0$ for $p_0 > 1/2$. Let $N \geq 2$. The even component has $\hat{a}_n = (1-d_n)(1-p_0)$, $\hat{b}_n = d_n(1-p_0) + (1-d_n)p_0$ and $\hat{c}_n = d_np_0$, so $\hat{a}_0 = 1-p_0$, $\hat{b}_0 = p_0$, and its normalization is $(1-d_n, (1-r)d_n, rd_n)$. Its profile d_n has geometric increments, so its Newton coefficients are proportional to $(q-1)^{k-1}$ and log-concave without internal zeros, and $0 < r \leq 1$. For the odd component, $\check{a}_n = (1-p_0)(1-d_{n+1})$, $\check{b}_n = (1-p_0)d_{n+1} + p_0(1-d_n)$ and $\check{c}_n = p_0d_n$. Geometric increments give $d_{n+1} - d_1 = qd_n$, so the normalization is

$$\Big(\frac{1-d_{n+1}}{1-d_1},\ \frac{(q-r)d_n}{1-d_1},\ \frac{rd_n}{1-d_1}\Big).$$

Its profile $qd_n/(1-d_1)$ has geometric increments with ratio q, and its downward coefficients are $r/q \leq 1$ times the profile. Theorem 3.1 applies to both components, and Theorem 4.1 concludes. Comparison with (2.4), using $d_n = [n]_q/[N]_q$ and $1 - d_1 = q[N-1]_q/[N]_q$, identifies the parameters. For $p_0 > 1/2$, $g(2,2;2) = \hat{g}(1,1;1)$ is the normalized diagonal coefficient $(1-r)d_1 < 0$ (for $N = 1$ see above). ∎

Theorem 4.3 *(Szwarc's quadratic system).* *Let $N \geq 1$ and $0 < p_0 \leq 1/2$. The array (4.2) has a symmetric positive orthogonality measure on $2N+1$ points, and for all $0 \leq m, n \leq 2N$*

$$R_m R_n = \sum_{k=0}^{2N} g(m,n;k) R_k, \qquad g(m,n;k) \geq 0, \qquad \sum_k g(m,n;k) = 1$$

on that support. If $1/2 < p_0 < 1$, then $g(2,2;2) < 0$.

Proof. Lemma 2.1 gives a positive measure on $2N+1$ points, and it is symmetric because $b \equiv 0$ (proof of Theorem 4.1). The array (4.2) satisfies the hypotheses of Corollary 4.2 with $c_{2n+1} = p_0$ and $d_n = n/N$, that is, with $q = 1$, which gives nonnegative linearization for $p_0 \leq 1/2$ and $g(2,2;2) < 0$ for $1/2 < p_0 < 1$. The coefficients sum to one because $R_k(1) = 1$ and 1 lies in the support, since $J\mathbf{1} = \mathbf{1}$. ∎

The even component of (4.2) is the Krawtchouk array $K_{N,r}$ with $r = p_0/(1-p_0)$, so that $Q_{2n}(y)$ is a Krawtchouk polynomial in y^2, as Szwarc notes, and its odd component is $K_{N-1,r}$. A second proof of Theorem 4.3, by a graded version of the algebra of [17], and explicit coefficient neighborhoods of (4.2) are given there.

5. Ratios given by weights

We use the notation of Section 3, extended as follows. The Newton coefficients of a sequence $(u_i)_{0 \leq i \leq N}$ are $\Delta^k u(0) = \sum_{i \leq k} (-1)^{k-i} \binom{k}{i} u_i$, so that $u_n = \sum_k \binom{n}{k} \Delta^k u(0)$. For a nonnegative, finitely supported sequence $(u_k)_{k \geq 0}$ that is log-concave without internal zeros in the sense of Lemma 3.3 we form $f_n(\ell) = f_n^u(\ell)$ and

$$W_{p,q}(\ell, a) = \sum_{n \geq p} \binom{n-p+q}{q} \binom{a}{n} f_n(\ell)$$

as in Section 3, and say from which sequence they are formed; without qualification, $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$ and $C_j = W_{2j+2,j+1}$ are formed from the Newton coefficients of the profile, as there. For $i \geq 1$ with $d_i > 0$ we write $\rho(i) = c_i/d_i$ for the ratio. Section 3 proves property (P) for constant ratios by expanding the product coefficients in a kernel $\omega_{\ell,a}$ on $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, which satisfies a recurrence in ℓ whose moves have weights formed from the $W_{p,q}$, and by matching each move of odd index with two-step detours through moves of even index (Lemmas 3.2 and 3.4). We extend that argument.

The kernel argument also works directly for normalized arrays, with independent diagonal and downward coefficients. Let $a_i + b_i + c_i = 1$ for $0 \leq i \leq N$, with $b_0 = c_0 = 0$, $a_N = 0$, $a_i > 0$ for $i < N$ and $c_i > 0$ for $i \geq 1$. Write

$$\kappa_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} c_i, \qquad \beta_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} b_i \qquad (1 \leq k \leq N),$$

with $\kappa_k = \beta_k = 0$ for $k = 0$ and $k > N$, and put $\varphi_n(\ell) = \sum_{h \geq 0} \binom{\ell}{h} \kappa_{n+h}$ and $\psi_n(\ell) = \sum_{h \geq 0} \binom{\ell}{h} \beta_{n+h}$, so that $\varphi_0(\ell) = c_\ell$ and $\psi_0(\ell) = b_\ell$ for $\ell \leq N$.

Theorem 5.1 *(Varying ratio).* *Let (κ_k) be log-concave without internal zeros and $\beta_k \geq 0$ for all k. Suppose that for $0 \leq \ell \leq N-1$*

$$\psi_1(\ell)\,\varphi_n(\ell+1) \geq c_{\ell+1}\,\psi_{n+1}(\ell) \quad (n \geq 2), \qquad \varphi_2(\ell)\,\psi_n(\ell+1) \geq c_{\ell+1}\,\psi_{n+2}(\ell) \quad (n \geq 1).$$

Then the system has nonnegative linearization.

Corollary 5.2. *If (κ_k) is log-concave without internal zeros and $\beta_k = w_k\kappa_k$ for $k \geq 1$ with $w_1 \geq w_2 \geq \cdots \geq 0$, then the system has nonnegative linearization. The ratio $c_i/(b_i+c_i)$ is nondecreasing in $i \geq 1$, and it is constant when w is constant on the support of κ; these are the arrays of Theorem 3.1.*

For example, if c is a convex quadratic in i ($\kappa_1, \kappa_2 > 0$, $\kappa_k = 0$ for $k \geq 3$) and $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$ with $w_1 > w_2 \geq 0$, scaled so that $b_N + c_N = 1$, the ratio $c_i/(b_i+c_i)$ increases strictly. The hypotheses exclude the counterexample recorded in Appendix C, whose diagonal coefficients $b = (0,1,1,1)/91$ have Newton coefficients of both signs. For $N \geq 2$, a strictly increasing linear ratio on a linear profile violates the corollary's assumptions because $\Delta^2 b(0) < 0$; Proposition 5.3 covers it when b is nondecreasing.

Proof of Theorem 5.1. Put $e_i = \prod_{j<i} a_j/c_{j+1}$ and $E_i = e_i R_i$. Then $xE_i = c_{i+1}E_{i+1} + b_iE_i + a_{i-1}E_{i-1}$ on the support. The out-of-range terms are omitted: at the endpoints, $xE_0 = c_1E_1$ and $xE_N = b_NE_N + a_{N-1}E_{N-1}$. The series $F = \sum_i E_i z^i$ satisfies $xF = \mathcal{L}_z F$ with

$$\mathcal{L}_z \;=\; z^{-1}c(\theta_z) + b(\theta_z) + z\big(1 - b(\theta_z) - c(\theta_z)\big),$$

where $c(\theta) = \sum_k \kappa_k\binom{\theta}{k}$ and $b(\theta) = \sum_k \beta_k\binom{\theta}{k}$. As in Section 3, $K_k = \sum_{m,n} L^k_{mn} z^m w^n$, with $E_mE_n = \sum_k L^k_{mn}E_k$, satisfies $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$, and $K_k = \sum \omega_{\ell,a} S^a Z^\ell$ with $\omega_{0,a} = \mathbf{1}_{a=k}$ and $\omega_{\ell,a} = 0$ for $\ell + a > N$. The computation in the proof of Lemma 3.2 gives

$$\frac{(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell}{z-w} \;=\; S^aZ^\ell + \sum_{p\geq 0}\binom{a}{p} S^{a-p}Z^\ell\Big(\varphi_p(\ell)(\chi_{p-2} - \chi_p) + \psi_p(\ell)(\chi_{p-1} - \chi_p)\Big),$$

with $\chi_{-2} = -Z^{-1}$ and $\chi_{-1} = 0$. Collecting terms as there, with $W_{p,q}$ formed from φ and $W^\psi_{p,q}$ from ψ, gives

$$\begin{aligned} c_{\ell+1}\omega_{\ell+1,a} \;=\; a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\geq 0}(-1)^j\Big(&A^\psi_j(\ell-j, a+2j+1)\,\omega_{\ell-j,a+2j+1} \\ &+ B_j(\ell-j, a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ &+ \big(C_j + C^\psi_j\big)(\ell-j-1, a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned}$$

where $A^\psi_j = W^\psi_{2j+1,j}$, $C^\psi_j = W^\psi_{2j+2,j+1}$, and $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$. The straight weight is $1 - c_{\ell+a} - b_{\ell+a} = a_{\ell+a}$, extended polynomially beyond N, and $\beta_0 = 0$. Up to this point no hypothesis of the theorem has been used: the recurrence holds for every normalized array, whatever the signs of its diagonal coefficients. Under the hypotheses, all weights are nonnegative on Ω.

For $j \ge 1$, $0 \le \ell \le N-1$ and $a \ge 0$, the following detour inequalities hold, with weights having a negative second argument interpreted as zero:

$$\begin{aligned} A_0^{\psi}(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}^{\psi}(\ell+1,a-2) &\ge 4c_{\ell+1}A_j^{\psi}(\ell,a), \\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4c_{\ell+1}B_j(\ell,a), \\ B_0(\ell,a)\big(C_{j-1}+C_{j-1}^{\psi}\big)(\ell+1,a-2) &\ge 4c_{\ell+1}\big(C_j+C_j^{\psi}\big)(\ell,a). \end{aligned}$$

The proof of Lemma 3.4 applies, with the lower bounds $A_0^{\psi}(\ell,a) \ge a\psi_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$. Lemma 3.3 for φ handles the products $\varphi_2(\ell)\varphi_n(\ell+1)$. The two hypotheses of the theorem handle $\psi_1(\ell)\varphi_n(\ell+1)$ for $n \ge 2$ and $\varphi_2(\ell)\psi_n(\ell+1)$ for $n \ge 1$, which are the indices that occur. The moves C_j and C_j^{ψ} have the same displacement and are treated as one move. With this, the proof of Theorem 3.1 gives $\omega \ge 0$, hence $L_{mn}^k \ge 0$ and $g(m,n;k) = L_{mn}^k e_k/(e_m e_n) \ge 0$. ∎

Proof of Corollary 5.2. Set $w_0 = w_1$; since $\kappa_0 = 0$, this leaves $\beta_0 = 0$. Fix ℓ and write $G_i = \varphi_i(\ell)$, $H_i = \psi_i(\ell)$ and $u_x = \binom{\ell}{x-i}$. For $i \ge 0$,

$$H_iG_{i+1} - H_{i+1}G_i = \sum_{x,y \ge i} u_x u_y \kappa_x \kappa_{y+1}(w_x - w_{y+1}).$$

Terms with $x \le y$ are nonnegative, and terms with $x = y+1$ vanish. The map $(x,y) \mapsto (y+1, x-1)$ is a bijection from the terms with $x \ge y+2$ onto those with $i+1 \le x \le y$; the remaining terms have $x = i$. A term and its image sum to $\kappa_x\kappa_{y+1}(w_{y+1} - w_x)(u_{y+1}u_{x-1} - u_x u_y) \ge 0$, because w is nonincreasing and the zero-extended binomial coefficients are log-concave. Hence $H_{i+1}G_i \le H_iG_{i+1}$.

Since the support of (G_i) is an interval and $\beta_k = 0$ where $\kappa_k = 0$, chaining gives $H_{n+1}G_1 \le H_1G_{n+1}$, $H_{n+2}G_n \le H_nG_{n+2}$ and $H_{n+2}G_{n+1} \le H_{n+1}G_{n+2}$. Lemma 3.3 with $(i,l) = (1,n)$ and $(2,n)$ reads $G_1(G_n + G_{n+1}) \ge CG_{n+1}$ and $G_2(G_n + G_{n+1}) \ge CG_{n+2}$, where $C = G_0 + G_1 = c_{\ell+1}$ and $G_n + G_{n+1} = \varphi_n(\ell+1)$. Since $G_1 \ge \kappa_1 > 0$, $H_1(G_n + G_{n+1}) \ge CH_1G_{n+1}/G_1 \ge CH_{n+1}$. If $G_{n+2} > 0$, then $G_2(H_n + H_{n+1}) \ge G_2H_{n+2}(G_n + G_{n+1})/G_{n+2} \ge CH_{n+2}$; if $G_{n+2} = 0$, then $H_{n+2} = 0$. These are the two hypotheses of Theorem 5.1. The case $i = 0$ of the first display, with $\beta_0 = \kappa_0 = 0$, reads $b_{\ell+1}c_\ell \le b_\ell c_{\ell+1}$; for $1 \le \ell < N$, division by $c_\ell c_{\ell+1} > 0$ shows that b_i/c_i is nonincreasing and $c_i/(b_i + c_i)$ is nondecreasing. If w is constant on the support of κ, then b is proportional to c, and with $d = b + c$ the array is that of Theorem 3.1. ∎

For arrays that are quadratic in the index, concave diagonal coefficients are allowed.

Proposition 5.3 *(Quadratic arrays).* *Let $b_n = \beta_1 n + \beta_2\binom{n}{2}$ and $c_n = \kappa_1 n + \kappa_2\binom{n}{2}$ $(0 \le n \le N)$ define an array as above. If $\kappa_2 \ge 0$, $\kappa_2 + \beta_2 \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, then the system has nonnegative linearization.*

Proof. In the recurrence of the proof of Theorem 5.1, which holds for every array, $\varphi_n = \psi_n = 0$ for $n \ge 3$. So all moves with $j \ge 1$ vanish, and the moves with $j = 0$ have the weights $A_0^{\psi}(\ell,a) = \sum_{n \ge 1}\binom{a}{n}\psi_n(\ell) = b_{\ell+a} - b_\ell$, $B_0(\ell,a) = \binom{a}{2}\kappa_2$ and $(C_0 + C_0^{\psi})(\ell,a) =$

$\binom{a}{2}(\kappa_2+\beta_2)$. These and the straight weights $a_{\ell+a}$ are nonnegative on Ω, so $\omega \geq 0$ by induction on ℓ, and $g(m,n;k) \geq 0$ as in the proof of Theorem 5.1. ■

With the linear profile $d_n = n/N$ and the ratio $c_n/d_n = r + \kappa n/N$ ($1 \leq n \leq N$, $r > 0$, $\kappa \geq 0$), one has $\kappa_2 = 2\kappa/N^2$ and $\kappa_2 + \beta_2 = 0$ for $N \geq 2$. For $N = 1$, property (P) follows directly from $R_1^2 = c_1 R_0 + b_1 R_1$ when $b_1 \geq 0$. So every linearly increasing ratio with a linear profile and nondecreasing b has property (P); b is nondecreasing exactly when $\kappa \leq (1-r)N/(2N-1)$. Exact computations [18] show that each of the three hypotheses is needed: dropping any one of them gives quadratic arrays with a negative coefficient at $N \leq 3$.

6. Linearly and quadratically increasing ratios

For curved profiles, linearly increasing ratios need not satisfy the hypotheses of Theorem 5.1: the Newton coefficients of b of order at least 3 can be negative, and then some moves of even index in the recurrence of its proof have negative weight and no two-step detour. A resummation of the positive moves handles this.

Theorem 6.1 *(Linearly increasing ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \leq 1$ and $\kappa \geq 0$, and put $\rho_i = r + \kappa i/N$, $a_i = 1 - d_i$, $b_i = (1-\rho_i)d_i$ and $c_i = \rho_i d_i$. If b is nondecreasing, then the system has nonnegative linearization. The condition that b be nondecreasing is equivalent to $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \leq N(1-r)$.*

For the linear profile $d_i = i/N$ the condition reads $\kappa \leq \frac{N}{2N-1}(1-r)$, and these are arrays of Proposition 5.3. For a convex profile $d_{N-1} \leq (N-1)/N$, so the admitted range of κ contains $[0, \frac{N}{2N-1}(1-r)]$. When $r < 1$, this containment is strict for a nonlinear profile. For example, $N = 4$, $d_i = (2^i - 1)/15$, $r = 1/2$ and $\kappa = 2/5$ are admitted, with $b = (0, 2/75, 3/50, 7/75, 1/10)$, although $(2N-1)\kappa/N = 7/10 > 1 - r$.

Proof. Write $s = 1 - r$ and $t = \kappa/N$, and let $f_n(\ell) = \Delta^n d(\ell)$ and $\nu_i = \Delta^i d(0)$ be as in Section 3, with $\nu_{-1} = \nu_0 = 0$. For $0 \leq i < N$ put $R_i = d_i/f_1(i)$; then $f_1(i) \geq \nu_1 > 0$, and log-concavity of $(f_n(i))_{n\geq 0}$ gives $R_{i+1} - R_i = \big(f_1(i)^2 - d_i f_2(i)\big)/\big(f_1(i)f_1(i+1)\big) \geq 0$. Since $b_{i+1} - b_i = f_1(i)\big(s - t(i+1) - tR_i\big)$ and the bracket is nonincreasing in i, b is nondecreasing exactly when its last increment is nonnegative, that is, when $t(N + R_{N-1}) \leq s$ with $R_{N-1} = d_{N-1}/(1-d_{N-1})$; this is the stated condition. In particular $tN \leq s$. Since $c_i = (r+ti)d_i$ and $b_i = (s-ti)d_i$, the Leibniz rule for differences gives, for $\ell \geq 0$ and $1 \leq n \leq N - \ell$, which includes every index occurring in the moves on Ω,

$$\varphi_n(\ell) = (r + t(\ell+n))f_n(\ell) + tnf_{n-1}(\ell), \qquad \psi_n(\ell) = h_n(\ell) - v_n(\ell),$$

with $h_n(\ell) = (s - t(\ell+n))f_n(\ell)$ and $v_n(\ell) = tnf_{n-1}(\ell)$; both are nonnegative when $\ell + n \leq N$. (Outside this range the formulas need not hold, because the Newton coefficients of b and c are set to zero beyond N.) Extend ν by zero outside $1 \leq i \leq N$ and put

$\widehat{\kappa}_i = p_i\nu_i + q_i\nu_{i-1}$ $(i \ge 1)$, $\widehat{\kappa}_0 = 0$, with $p_i = r + ti$ and $q_i = ti$; the Newton sequence κ of c is the truncation of $\widehat{\kappa}$ at N. Then

$$\begin{aligned}\widehat{\kappa}_i^2 - \widehat{\kappa}_{i-1}\widehat{\kappa}_{i+1} &= p_i^2(\nu_i^2 - \nu_{i-1}\nu_{i+1}) + q_i^2(\nu_{i-1}^2 - \nu_{i-2}\nu_i)\\ &\quad + q_{i-1}p_{i+1}(\nu_i\nu_{i-1} - \nu_{i-2}\nu_{i+1}) + t^2(\nu_{i-1}\nu_{i+1} + \nu_{i-2}\nu_i + 2\nu_i\nu_{i-1}),\end{aligned}$$

and each term is nonnegative by log-concavity of ν. The positive support of $\widehat{\kappa}$ is that of ν when $t = 0$ and extends it by one index when $t > 0$, an interval in both cases. Truncation at N preserves log-concavity and interval support, so κ is log-concave without internal zeros, and Lemma 3.3 applies to f and to φ.

Moves. We use the recurrence of the proof of Theorem 5.1, which holds for every array. For $j \ge 1$ split $A_j^{\psi} = H_j - V_j$, where $H_j = W^h_{2j+1,j}$ and $V_j = W^v_{2j+1,j}$ are formed from h_n and v_n in place of ψ_n; both are nonnegative on Ω. Keep $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ as one move; it is nonnegative because b is nondecreasing. The weights B_j, formed from φ, and $C_j + C_j^{\psi} = W^f_{2j+2,j+1}$, formed from f, are nonnegative. Write $C'_j = C_j + C_j^{\psi}$.

Detours. All detour inequalities below are asserted for source nodes $(\ell, a) \in \Omega$; in the mixed products the occurring indices satisfy $n \le a - 2$, hence $\ell + n + 2 \le N$. Since $\varphi_2(\ell) \ge (r + t(\ell+1))f_2(\ell)$ and $c_{\ell+1} = (r + t(\ell+1))d_{\ell+1}$, Lemma 3.3 for f gives

$$\begin{gathered}\varphi_2(\ell)f_n(\ell+1) \ge c_{\ell+1}f_{n+2}(\ell), \qquad \varphi_2(\ell)h_n(\ell+1) \ge c_{\ell+1}h_{n+2}(\ell),\\ \varphi_2(\ell)v_n(\ell+1) \ge \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell) \quad (n \ge 2),\end{gathered}$$

using $s - t(\ell + n + 1) \ge s - t(\ell + n + 2) \ge 0$ for the second and Lemma 3.3 with indices $(2, n-1)$ for the third. As in the proof of Lemma 3.4, with $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell)$, for $0 \le \ell \le N - 1$ and second factors at $(\ell + 1, a - 2)$,

$$\begin{aligned} B_0B_{j-1} &\ge 4c_{\ell+1}B_j \ (j \ge 1), & B_0H_{j-1} &\ge 6c_{\ell+1}H_j \ (j \ge 2),\\ B_0V_{j-1} &\ge 6c_{\ell+1}V_j \ (j \ge 2), & B_0C'_{j-1} &\ge 6c_{\ell+1}C'_j \ (j \ge 1).\end{aligned}$$

The first uses Lemma 3.3 for φ. The coefficient ratios are, on $m \ge 2j+2$, $2j+1$, $2j+1$ and $2j+2$ respectively,

$$\frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{jm(m-1)}{2(m-j-1)}, \qquad \frac{j(m-1)(m-2)}{2(m-j-1)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)};$$

they increase with j and are at least 5, 10, 6 and 6. Finally,

$$A_0^{\psi}(\ell, a)B_0(\ell+1, a-1) + B_0(\ell, a)A_0^{\psi}(\ell+1, a-2) \ge c_{\ell+1}\big(4H_1(\ell, a) - 2V_1(\ell, a)\big). \quad (6.1)$$

Only $a \ge 3$ needs proof. Put $u_n = f_n(\ell)$, $y_n = \binom{a}{n}u_n$, $D = u_0 + u_1 = d_{\ell+1}$, $\alpha = s - t(\ell + a)$,

$$\begin{aligned} X &= \sum_{n\ge1} y_n, \qquad Y = \sum_{n\ge2} y_n, \qquad J = \sum_{n=2}^{a-1}\tbinom{a-1}{n}u_n,\\ X' &= \sum_{j\ge1}\tbinom{a-2}{j}(u_j + u_{j+1}), \qquad Y' = \sum_{j\ge2}\tbinom{a-1}{j}(u_j + u_{j+1}),\end{aligned}$$

so that $X = au_1 + Y$, $X' = (a-2)u_1 + J$ and $Y' = Y - (a-1)u_2$. Then $A_0^{\psi}(\ell, a) = \alpha X - tau_0$ and $A_0^{\psi}(\ell+1, a-2) = (\alpha+t)X' - t(a-2)D$; both are nonnegative, and the two weights B_0 are at least $(r + t(\ell+1))Y'$ and $(r+t(\ell+1))Y$. With $H_* = \sum_{n\geq 3}(n-2)y_n$ and $T = \sum_{n\geq 2}(n-3)(a-n)y_n$ one has $H_1 = \sum_{n\geq 3}(n-2)(\alpha + t(a-n))y_n$ and, after a shift of index, $V_1 = t\sum_{n\geq 2}(n-1)(a-n)y_n$, so $4H_1 - 2V_1 = 4\alpha H_* + 2tT$. Since $c_{\ell+1} = (r + t(\ell+1))D$, it suffices to show

$$(\alpha X - tau_0)Y' + Y\big((\alpha+t)X' - t(a-2)D\big) \;\geq\; D(4\alpha H_* + 2tT). \tag{6.2}$$

The coefficient of α in the difference of the two sides is $K = XY' + YX' - 4DH_*$. By Lemma 3.3, $u_i(u_j + u_{j+1}) \geq Du_{i+j}$ for $i, j \geq 1$, so $XY' \geq au_1Y' \geq D\sum_{n\geq 3} ny_n$ and $YX' \geq \binom{a}{2}u_2X' \geq D\sum_{n\geq 3}\binom{n}{2}y_n$; hence $K \geq \frac{D}{2}\sum_{n\geq 3}\big((n-3)(n-4)+4\big)y_n \geq 0$, and (6.2) holds when $t = 0$. Let $t > 0$. Monotonicity of b at $\ell + a - 1$ gives $\alpha \geq tR_{\ell+a-1} \geq tR_\ell = tu_0/u_1$, so it suffices to prove (6.2) at $\alpha = tu_0/u_1$. Both sides are homogeneous of degree two in (u_n); take $u_1 = 1$ and write $z = u_0$ and $q = u_2$. If $q = 0$, then $u_n = 0$ for $n \geq 2$ and both sides vanish. Otherwise $0 \leq z \leq 1/q$ by log-concavity. At $\alpha = tz$ the two weights A_0^{ψ} are tzY and $t(z+1)J$, and (6.2) divided by t reads $F(z) \geq 0$, where

$$\begin{aligned} F(z) \;&=\; zYY' + (z+1)YJ - (z+1)(4zH_* + 2T) \\ &=\; (1-qz)F(0) + qzF(1/q) + 4H_*z(1/q - z), \end{aligned}$$

the second form holding because F is quadratic with leading coefficient $-4H_*$. It remains to show $F(0) \geq 0$ and $F(1/q) \geq 0$. Put $L_3 = \frac{3}{2}$, $L_n = \frac{n(n+2)}{6}$ for $n \geq 4$, and $E_n = \frac{2(n^2-5)}{n+1}$. In YJ keep, for each n, the product with indices $(2,2)$ if $n = 3$, and the products with indices $(2, n-1)$ and $(3, n-2)$ if $n \geq 4$; by log-concavity with $u_1 = 1$, $u_iu_{n+1-i} \geq u_n$, and $\binom{a-1}{j} \geq \binom{a-i}{j}$, so $YJ \geq \sum_{n=3}^{a-1} L_n(a-n)y_n$. In Y keep the term $\binom{a}{2}q$; as $q(u_{n-1} + u_n) \geq (q+1)u_n$ for $n \geq 3$, $YY' \geq (q+1)\sum_{n=3}^{a}\frac{n(a-1)}{2}y_n$. Since $L_n \geq 2(n-3)$,

$$F(0) \;=\; YJ - 2T \;\geq\; 2(a-2)y_2 + \sum_{n=3}^{a-1}\big(L_n - 2(n-3)\big)(a-n)y_n \;\geq\; 0.$$

Using $u_n \leq qu_{n-1}$ and a shift of index, $\frac{4}{q}H_* + 2T \leq \sum_{n=2}^{a-1} E_n(a-n)y_n$, where $E_2 < 0$. As $L_n + \frac{n}{2} \geq E_n$ for $n \geq 3$,

$$\frac{F(1/q)}{1+1/q} \;=\; YJ + \frac{YY'}{q+1} - \frac{4}{q}H_* - 2T \;\geq\; \sum_{n=3}^{a-1}\Big(L_n + \frac{n}{2} - E_n\Big)(a-n)y_n \;\geq\; 0.$$

The numerical facts used are $L_3 = \frac{3}{2} \geq 0$, $L_3 + \frac{3}{2} = 3 \geq E_3 = 2$, and, for $n \geq 4$, $6(L_n - 2(n-3)) = (n-5)^2 + 11$ and $6(n+1)(L_n + \frac{n}{2} - E_n) = n^3 - 6n^2 + 5n + 60 > 0$.

Resummation. Regard the moves as operators on functions on Ω, each divided by c at the destination level; all strictly raise ℓ, and inequalities between operators are entrywise. Let P and Q be the sums of the B_j with j even and odd, $C = C_+ - C_-$ with $C_\pm$ the sums of the C'_j with j even and odd, H_+ and H_- the sums of the H_j with $j \geq 2$ even and $j \geq 3$ odd, V_+ and V_- the sums of the V_j with j odd and $j \geq 2$ even, A_0 the operator of

A_0^{ψ}, and U the straight moves. The recurrence reads $\omega = \beta + (U + P - Q + C + A)\omega$ with $A = A_0 + H_+ - H_- - H_1 + V_+ - V_-$ and $\beta = \omega|_{\ell=0} \geq 0$. The inequalities above give

$$P^2 \geq 4Q, \quad H_+P \geq 6H_-, \quad V_+P \geq 6V_-, \quad C_+P \geq 6C_-, \quad A_0P + PA_0 + 2V_1 \geq 4H_1.$$

Let $G = (I - P/2)^{-1} = \sum_{m=0}^{N}(P/2)^m \geq 0$. Then $PG = GP = 2(G - I)$ and $(I - P/6)G = \frac{2}{3}G + \frac{1}{3}I$, so $G(H_+ - H_-)G \geq GH_+(I - P/6)G \geq 0$, $GCG \geq 0$ and $G(V_+ - V_-)G \geq \frac{2}{3}GV_1G$. Hence

$$\begin{aligned} GAG &\geq G\big(A_0 - H_1 + \tfrac{2}{3}V_1\big)G \geq G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G \\ &= \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \geq 0. \end{aligned}$$

With $R = U + P^2/4 - Q \geq 0$, the operator $M = G(R + C + A)G$ is nonnegative and strictly raises ℓ. Since $(I - P/2)^2\omega = \beta + (R + C + A)\omega$, the function $v = (I - P/2)\omega$ satisfies $v = G\beta + Mv$. Therefore $\omega = G\sum_{q=0}^{N} M^qG\beta \geq 0$, and $g(m, n; k) \geq 0$ as in the proof of Theorem 5.1. ■

Theorem 6.2 *(Convex quadratic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \geq 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 6.1 ($\tau = 0$). As in the proof of that theorem, $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i + 1) - (t + \tau i)R_i\big)$ with a nonincreasing bracket, so b is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N - 1)R\big) \leq 1 - r$ with $R = d_{N-1}/(1 - d_{N-1})$. The Newton coefficients of c need not be log-concave: for $d_i = i/4$ and $\rho(x) = \frac{1}{2} + \binom{x}{2}/40$ they are $(1/8, 1/80, 3/160, 0)$. The proof below replaces their log-concavity by a weighted detour inequality.

Proof. Write $f_n(\ell) = \Delta^n d(\ell)$, $\Delta\rho(x) = t + \tau x$ and $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 6.1. Since $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i + 1) - \Delta\rho(i)R_i\big)$ and b is nondecreasing, $1 - \rho(m + 1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and $\rho(i) \leq 1$ for $i \leq N$. For $\ell + n \leq N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell + n)f_n + n\Delta\rho(\ell + n - 1)f_{n-1} + \binom{n}{2}\tau f_{n-2}, \qquad \psi_n = h_n - v_n - w_n,$$

with $h_n = (1 - \rho(\ell + n))f_n$, $v_n = n\Delta\rho(\ell + n - 1)f_{n-1}$ and $w_n = \binom{n}{2}\tau f_{n-2}$; here φ_n, h_n, v_n and w_n are nonnegative, while ψ_n need not be.

Moves. In the recurrence of the proof of Theorem 5.1, keep $A_0^{\psi} = b_{\ell+a} - b_\ell \geq 0$ and split $A_j^{\psi} = H_j - V_j - W_j$ for $j \geq 1$, where H_j, V_j and W_j are $W_{2j+1,j}$ formed from h_n, v_n and w_n; the weights B_j, formed from φ, and $C_j' = C_j + C_j^{\psi} = W_{2j+2,j+1}^{f}$ are nonnegative. With the signs $(-1)^j$, the moves W_j behave like the V_j.

Detours of H, V, W and C'. All detours are asserted for sources $(\ell, a) \in \Omega$, with second factors at $(\ell + 1, a - 2)$, marked by a prime. Since $\varphi_2(\ell) \geq \rho(\ell + 1)f_2(\ell)$ and $c_{\ell+1} =$

$\rho(\ell+1)d_{\ell+1}$, Lemma 3.3 for f gives, for $n \le a-2$,

$$\varphi_2 f'_n \ge c_{\ell+1} f_{n+2}, \qquad \varphi_2 h'_n \ge c_{\ell+1} h_{n+2},$$
$$(n+1)(n+2)\varphi_2 v'_n \ge n^2 c_{\ell+1} v_{n+2}, \qquad (n+1)(n+2)\varphi_2 w'_n \ge n(n-1) c_{\ell+1} w_{n+2},$$

the last two for $n \ge 3$. The second uses $1-\rho(\ell+n+1) \ge 1-\rho(\ell+n+2) \ge 0$; the third uses Lemma 3.3 with indices $(2, n-1)$ and $(n+1)\Delta\rho(\ell+n) - n\Delta\rho(\ell+n+1) = t+\tau\ell \ge 0$; the fourth uses indices $(2, n-2)$. As in the proof of Lemma 3.4, with $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$ and $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$,

$$B_0 H'_{j-1} \ge 10 c_{\ell+1} H_j, \qquad B_0 V'_{j-1} \ge \tfrac{9}{2} c_{\ell+1} V_j, \qquad B_0 W'_{j-1} \ge 3 c_{\ell+1} W_j \quad (j \ge 2),$$
$$B_0\, C'_{j-1}(\ell+1, a-2) \ge 6 c_{\ell+1} C'_j \quad (j \ge 1).$$

The coefficient ratios are $\frac{jm(m-1)}{2(m-j-1)}$, $\frac{j(m-2)^2}{2(m-j-1)}$ and $\frac{j(m-2)(m-3)}{2(m-j-1)}$ on $m \ge 2j+1$, and $\frac{(j+1)m(m-1)}{2(m-j-1)}$ on $m \ge 2j+2$; they increase with j, and at the smallest j the bounds are $(m-5)(m-6) \ge 0$, $(m-5)(2m-7) \ge 0$, $m \ge 5$ and $(m-3)(m-4) \ge 0$.

Detour of B. We show, for $n \ge 2$ and $\ell+n+2 \le N$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \ge 8(n-1)\,c_{\ell+1}\varphi_{n+2}(\ell). \tag{6.3}$$

Granting this, $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$, (6.3) with $n = m-2$, $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$ and $\binom{m}{2}\frac{8(m-3)}{m(m-1)} = 4(m-3)$ give

$$\begin{aligned} B_0(\ell,a) B_{j-1}(\ell+1, a-2) &\ge 4c_{\ell+1} \sum_{m \ge 2j+2} (m-3)\binom{m-j-3}{j-1}\binom{a}{m}\varphi_m(\ell) \\ &\ge 4 c_{\ell+1} B_j(\ell,a) \end{aligned}$$

for $j \ge 1$, because $(m-3)\binom{m-j-3}{j-1} - \binom{m-j-2}{j} = \binom{m-j-3}{j-1}\frac{(j-1)(m-2)}{j} \ge 0$. (With constant 1 in place of $\frac{(n+1)(n+2)}{8(n-1)}$, (6.3) can fail: for $N = 20$, $d_i = (2^i-1)/(2^{20}-1)$, $r = \frac12$, $t = 0$, $\tau = 10^{-3}$ one gets $\varphi_2(16)\varphi_2(17) < c_{17}\varphi_4(16)$.)

To prove (6.3), put $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$, and $p = \rho(\ell+1)$, $K_n = (n+1)(n+2)$, $L_n = 8(n-1)$. Both sides are homogeneous of degree two in u; take $u_1 = 1$ and write $z = u_0$, $q = u_2$. By convexity and log-concavity, $0 \le z \le \ell$ and $zq \le 1$. If $q = 0$ or $u_n = 0$, then $\varphi_{n+2}(\ell) = 0$. Otherwise put $y = u_{n+1}/u_n \in [0, q]$. Log-concavity gives $u_{n+2} \le y^2 u_n$, and for $n \ge 3$ also $u_{n-1} \ge u_n/q$ and $u_{n-2} \ge u_n/q^2$. Hence $\varphi_n(\ell+1) \ge u_n \mathcal{L}(y)$ and $\varphi_{n+2}(\ell) \le u_n \mathcal{T}(y)$, where

$$\mathcal{T}(y) = \rho(\ell+n+2)y^2 + (n+2)\Delta\rho(\ell+n+1)y + \binom{n+2}{2}\tau,$$

$\mathcal{L}(y) = \rho(\ell+n+1)(1+y) + n\Delta\rho(\ell+n)(1+q^{-1}) + \binom{n}{2}\tau(q^{-1}+q^{-2})$ for $n \ge 3$, and $\mathcal{L}(y) = \rho(\ell+3)(1+y) + 2\Delta\rho(\ell+2)(1+q^{-1}) + \tau(z+1)q^{-1}$ for $n = 2$ (exact). In both cases $\mathcal{L}(y) = A + By$ with $A \ge B \ge 0$, so $\mathcal{L}(0) \ge \mathcal{L}(q)/(1+q)$; and $\mathcal{T}(q) \ge (1+q)\mathcal{T}(0)$ because $(n+2)\Delta\rho(\ell+n+1) \ge (n+2)(n+1)\tau$. The function $\mathcal{F}(y) = K_n\varphi_2(\ell)\mathcal{L}(y) -$

$L_n p(z+1)\mathcal{T}(y)$ is concave with $\mathcal{F}(0) \geq \mathcal{F}(q)/(1+q)$, so it suffices that $\mathcal{F}(q) \geq 0$, which is (6.3) for the configuration $u_0 = z$, $u_k = q^{k-1}$ $(k \geq 1)$.

Next we reduce to $z = \min(\ell, 1/q)$. For $n \geq 3$, only $\varphi_2(\ell) = A_2 + \tau z$ and $c_{\ell+1} = p(z+1)$ depend on z, where $A_2 = \rho(\ell+2)q + 2\Delta\rho(\ell+1) \geq \tau$, so $\varphi_2(\ell)/(z+1)$ is nonincreasing in z. For $n = 2$, write also $\varphi_2(\ell+1) = G_2 + \tau(z+1)$ with $G_2 = \rho(\ell+3)q(1+q) + 2\Delta\rho(\ell+2)(1+q)$. For $z \leq Z \leq \ell$ the quotient $\varphi_2(\ell)\varphi_2(\ell+1)/(z+1)$ at z minus its value at Z equals $(Z-z)\big(G_2(A_2-\tau)/((z+1)(Z+1)) - \tau^2\big) \geq 0$, since $G_2(A_2-\tau) \geq 2(\ell+2)(2\ell+1)\tau^2 \geq (\ell+1)^2\tau^2$.

For the remaining configurations we use the following polynomial inequality. For an integer $n \geq 2$ and reals $h \geq 1$, $p, \tau \geq 0$ and $s \geq (h-1)\tau$, put $A = p + (2h-1)s + h^2\tau$, $B = p + nh(s+\tau) + \binom{n}{2}h^2\tau$ and $C = p + ((n+2)h-1)s + \binom{n+2}{2}h^2\tau$. Then

$$K_n AB - L_n pC \geq 0. \tag{6.4}$$

Indeed, $K_nAB - L_npC$ is nondecreasing in s (its ps-coefficient is $a_n((n+2)h-1)$ with $a_n = n^2 - 5n + 10 > 0$, and its other s-dependent coefficients are nonnegative), and at $s = (h-1)\tau$ it equals $a_np^2 + C_n(h)p\tau + D_n(h)\tau^2$, where $C_n(h) = \frac{1}{2}\big(h^2U_n + h(10n^2+14n-60) + 2a_n\big)$, $U_n = n^4 - 4n^3 - 21n^2 + 12n + 60$ and $D_n(h) = \frac{1}{2}h^2n(n+1)^2(n+2)(3h^2-3h+1)$. For $n \geq 7$ all three coefficients are nonnegative, since $U_n = k^4 + 24k^3 + 189k^2 + 502k + 144$ with $k = n-7$. For $2 \leq n \leq 6$, $4a_nD_n - C_n^2$ is a polynomial in $v = h-1$ with positive coefficients:

n	v^0	v^1	v^2	v^3	v^4
2	576	2880	5616	4992	1664
3	1520	6240	9744	7200	2160
4	6300	27000	42660	30600	8604
5	25200	126000	229500	184800	56000
6	62720	376320	774144	677376	216576

so $a_np^2 + C_np\tau + D_n\tau^2 = a_n\big(p + \frac{C_n\tau}{2a_n}\big)^2 + \frac{4a_nD_n - C_n^2}{4a_n}\tau^2 \geq 0$.

If $1/q \leq \ell$, take $z = 1/q$, $h = 1 + 1/q$ and $s = t + \tau\ell \geq (h-1)\tau$. For the configuration above, $\mathcal{F}(q)$ divided by $q(1+q)$ equals $K_nAB - L_npC$, which is nonnegative by (6.4).

If $\ell < 1/q$, take $z = \ell$, and write $w = 1/q - \ell > 0$, $h_0 = \ell + 1$, $h = h_0 + w$ and $s = t + \tau\ell$. With $A(w) = p + (2h-1)s + \tau\big((\ell+2)h - \ell - 1\big)$ and $B(w)$, $C(w)$ as above (with this h), $\mathcal{F}(q)$ divided by q^2 equals $\mathcal{E}_n(w) = K_n(h_0+w)A(w)B(w) - L_nh_0pC(w)$ for $n \geq 3$, and $\mathcal{E}_2(w) - 12A(w)\tau w(\ell+w)^2$ for $n = 2$. At $w = 0$ both equal $h_0(K_nAB - L_npC)$ with $h = h_0$, which is nonnegative by (6.4). Expanding in w, with $\bar{A} = A(0) - p$ and $\bar{B} = B(0) - p$, every coefficient of w, w^2, w^3 and w^4 is a sum of nonnegative terms: for $n \geq 3$,

$$\begin{aligned}
[w] &= K_np^2 + p(a_1t + b_1\tau) + K_n\big(\bar{A}\bar{B} + h_0(\eta\bar{B} + \bar{A}\theta)\big),\\
[w^2] &= p(a_2t + b_2\tau) + K_n\big(\eta\bar{B} + \bar{A}\theta + h_0(\eta\theta + \bar{A}\epsilon)\big),\\
[w^3] &= K_n\big(\eta\theta + A(0)\epsilon + h_0\eta\epsilon\big), \qquad [w^4] = K_n\eta\epsilon,
\end{aligned}$$

where

$$\begin{aligned}
\eta &= 2s + (\ell+2)\tau, \qquad \theta = n(s+\tau) + n(n-1)\tau h_0, \qquad \epsilon = \tbinom{n}{2}\tau,\\
a_1 &= (n+2)\big(2\ell(n^2-n+6) + 2n^2 - 3n + 11\big), \qquad a_2 = K_n(n+2),\\
b_1 &= \tfrac{n+2}{2}(\ell^2P_n + \ell Q_n + R_n), \qquad b_2 = \tfrac{K_n}{2}\big(3n(n-3)(\ell+1) + 14\ell + 12\big),
\end{aligned}$$

with $P_n = 3k^3+15k^2+6k+8$, $Q_n = 6k^3+30k^2+20k+16$ and $R_n = 3k^3+15k^2+16k+16$ for $k = n-3$; for $n = 2$ the coefficients of w, w^2 and w^3 are

$$\begin{aligned}
&12p^2 + 4(16\ell+13)pt + 8(\ell+1)(2\ell+3)p\tau + 48(\ell+1)(3\ell+2)t^2\\
&\qquad + 12(36\ell^3+92\ell^2+76\ell+21)t\tau + 12(27\ell^4+93\ell^3+114\ell^2+62\ell+13)\tau^2,\\
&48pt + 12(2\ell+3)p\tau + 24(6\ell+5)t^2\\
&\qquad + 12(36\ell^2+67\ell+29)t\tau + 12(27\ell^3+78\ell^2+67\ell+19)\tau^2,\\
&12\big(2s+(\ell+2)\tau\big)\big(2t+(3\ell+5)\tau\big).
\end{aligned}$$

This proves (6.3).

Key inequality. We show, for $(\ell,a)\in\Omega$ with $a\ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^{\psi}(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1-2V_1-W_1\big)(\ell,a). \quad (6.5)$$

Use the notation of the proof of Theorem 6.1 ($u_n = f_n(\ell)$, y_n, D, X, Y, X', Y', J, H_*, T), and put $\alpha = 1-\rho(\ell+a)$, $s = \Delta\rho(\ell+a-1)$, $p = \rho(\ell+1)$ and $Z = \sum_{n\ge2}\binom{a-n}{2}y_n$. Then $\alpha \ge sR_{\ell+a-1} \ge su_0/u_1$ and $s\ge\tau(a-1)$. Backward differences give

$$\begin{aligned}
A_0^{\psi}(\ell,a) &= \alpha X - sau_0 + \tau\tbinom{a}{2}u_0,\\
A_0^{\psi}(\ell+1,a-2) &= (\alpha+s)X' - s(a-2)D + \tau\tbinom{a-1}{2}D,
\end{aligned}$$

both nonnegative, and $B_0(\ell,a)\ge pY$, $B_0(\ell+1,a-1)\ge pY'$, $c_{\ell+1} = pD$. Since $1-\rho(\ell+n) = \alpha+(a-n)s-\tau\binom{a-n}{2}$, reindexing gives $4H_1-2V_1 = 4\alpha H_* + 2sT + 4\tau Z$ and $W_1 = \tau\sum_{n\ge1} n\binom{a-n}{2}y_n$. The argument of the proof of Theorem 6.1 with s in place of t, keeping the surplus of the two endpoints ($F(0)\ge 2(a-2)y_2$ and $F(1/q)\ge\frac23(1+1/q)(a-2)y_2$, from the term $n=2$ and $E_2=-\frac23$), shows

$$(\alpha X - sau_0)Y' + Y\big((\alpha+s)X' - s(a-2)D\big) - D(4\alpha H_* + 2sT) \ge \tfrac23 sD(a-2)y_2$$

for $\alpha u_1\ge su_0$; the right side of this bound is linear in $z = u_0/u_1$ and is subtracted before the interpolation. With $s\ge\tau(a-1)$, the difference of the two sides of (6.5) divided by p is therefore at least

$$\begin{aligned}
\tau\Big(\tbinom{a}{2}u_0Y' + D\Big[\tbinom{a-1}{2}y_1 + \tfrac{(a-2)(a+11)}{6}y_2 + (2a-5)y_3\\
+ \sum_{n\ge4}\Big(\tbinom{a-1}{2} + (n-4)\tbinom{a-n}{2}\Big)y_n\Big]\Big) \ge 0.
\end{aligned}$$

Resummation. With $W_\pm$ the sums of the W_j with j odd and $j\ge2$ even, the recurrence reads as in the proof of Theorem 6.1 with $A = A_0 - H_1 + (H_+-H_-) + (V_+-V_-) + (W_+-W_-)$, and the inequalities above give $P^2\ge 4Q$, $H_+P\ge 10H_-$, $V_+P\ge\frac92 V_-$, $W_+P\ge 3W_-$, $C_+P\ge 6C_-$ and $A_0P+PA_0+2V_1+W_1\ge 4H_1$. Since $(I-P/\mu)G = (1-\frac2\mu)G+\frac2\mu I$, one has $G(H_+-H_-)G\ge 0$, $GCG\ge 0$, $G(V_+-V_-)G\ge\frac59 GV_1G$ and $G(W_+-W_-)G\ge\frac13 GW_1G$, hence

$$\begin{aligned}
GAG &\ge G\big(A_0 - \tfrac14(A_0P+PA_0) + \tfrac{1}{18}V_1 + \tfrac{1}{12}W_1\big)G\\
&= \tfrac12(GA_0+A_0G) + \tfrac{1}{18}GV_1G + \tfrac{1}{12}GW_1G \ge 0.
\end{aligned}$$

With $U+P^2/4-Q\ge 0$ this gives $M\ge 0$, and (P) follows as before. ∎

7. Ratios of higher degree

For profiles of Newton degree at most two the ratio can have any degree, provided the Newton coefficients of c are nonnegative and log-concave without internal zeros; no key inequality is needed.

Theorem 7.1 *(Low-degree profiles).* *Let $d_i = \nu_1 i + \nu_2 \binom{i}{2}$ with $\nu_1 > 0$, $\nu_2 \geq 0$ and $d_N = 1$, let $c_0 = 0$, and suppose that the Newton coefficients $\kappa_k = \Delta^k c(0)$, $1 \leq k \leq N$, are log-concave without internal zeros. Put $a_i = 1 - d_i$ and $b_i = d_i - c_i$. If b is nondecreasing, then the system has nonnegative linearization.*

Proof. We use the recurrence of the proof of Theorem 5.1, which holds for every array, with $\varphi_n(\ell) = \Delta^n c(\ell)$, $\psi_n(\ell) = \Delta^n b(\ell)$ and $f_n(\ell) = \Delta^n d(\ell)$ for the polynomial extensions. Since $f_n = 0$ for $n \geq 3$, every move A_j^{ψ} with $j \geq 1$, which involves only $n \geq 2j+1 \geq 3$, equals $-\widetilde{B}_j$ with $\widetilde{B}_j = W_{2j+1,j}$ formed from φ; the moves $C_j + C_j^{\psi} = W_{2j+2,j+1}^{f}$ vanish for $j \geq 1$, and $C_0 + C_0^{\psi} = \binom{a}{2}\nu_2 \geq 0$. The weight $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ is nonnegative on Ω, as are the straight weights and all B_j and $\widetilde{B}_j$, since $(\varphi_n(\ell))_n$ is obtained from (κ_k) by the shifts of Lemma 3.3 and is therefore nonnegative and log-concave without internal zeros. Lemma 3.3 for φ gives, as in Lemma 3.4, $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}B_j(\ell, a)$ for $j \geq 1$, and with $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell)$ also $B_0(\ell, a)\widetilde{B}_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}\widetilde{B}_j(\ell, a)$ for $j \geq 2$, because the coefficient quotients $jm(m-1)/(2(m-j-1))$ on $m \geq 2j+1$ are at least 10. In the resummation of the proof of Theorem 6.1 take $H = 0$ and $V_j = \widetilde{B}_j$; the key inequality is then trivial, and $M \geq 0$ gives (P). ∎

Corollary 7.2. *In Theorem 7.1 let $c_i = \rho(i)d_i$ with $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\geq 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Then the Newton coefficients of c are log-concave without internal zeros. Hence every such ratio, of any degree, with nondecreasing b gives nonnegative linearization on these profiles.*

Proof. Put $s_k = \rho_k + \rho_{k-1}$ ($\rho_{-1} = 0$), which is log-concave without internal zeros, and $h = \nu_2/2$. The Leibniz rule gives $\kappa_k = k e_k$ for $k \geq 1$, with $e_k = (\nu_1 + h(k-1))s_k + h(k-1)s_{k-1}$. Write $\alpha = \nu_1 + h(k-1)$ and $\beta = h(k-1)$; then

$$\begin{aligned} e_k^2 - e_{k-1}e_{k+1} \;=\; & \alpha^2(s_k^2 - s_{k-1}s_{k+1}) + \beta^2(s_{k-1}^2 - s_{k-2}s_k) \\ & + (\beta - h)(\alpha + h)(s_{k-1}s_k - s_{k-2}s_{k+1}) \\ & + h^2(s_{k-1}s_{k+1} + s_{k-2}s_k + 2s_{k-1}s_k), \end{aligned}$$

and every term is nonnegative for $k \geq 2$ by log-concavity of s. So $\kappa_k^2 \geq k^2 e_{k-1}e_{k+1} \geq \kappa_{k-1}\kappa_{k+1}$. If $\kappa_k = 0$ with $k \geq 1$, then $s_k = 0$, so $\rho_k = 0$; as $\rho_0 > 0$ and ρ has no internal zeros, $\rho_m = 0$ for $m \geq k$, and $\kappa_{k+1} = 0$. ∎

For example, on the linear profile $d_i = i/N$ the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ with $r > 0$ and $\varepsilon \geq 0$, whose Newton coefficients $r\varepsilon^k$ are geometric, give nonnegative linearization whenever b is nondecreasing. Nonnegativity of the ratio's Newton coefficients alone does not suffice: the cubic counterexample in Appendix C has the linear profile $d_i = i/4$

and the ratio with Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$, which are nonnegative and satisfy the log-concavity inequalities but have internal zeros. Theorem 6.2 and Corollary 7.2 are not comparable: the former admits convex quadratic ratios with $t^2 < r\tau$ on every log-concave Newton profile, the latter ratios of every degree on profiles of Newton degree at most two.

Products of Newton sequences. The proof of Corollary 7.2 used only that the Newton coefficients of $c = \rho d$ are log-concave. This holds for every profile.

Proposition 7.3 *(Products).* *Let $(\rho_i)_{i\ge 0}$ and $(\nu_j)_{j\ge 0}$ be nonnegative and log-concave without internal zeros, extended by zero at negative indices, and let $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$ for $x \in \mathbb{Z}_{\ge 0}$, where the sums are finite. Then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of fg are nonnegative and log-concave without internal zeros.*

The proof, which is self-contained and works in the Newton basis, is given in Appendix D.

In the language of random sets: if S and T are independent random subsets of $[n]$ whose laws are invariant under permutations and whose cardinalities are ultra-log-concave of order n in the sense of Liggett [12], then so is $|S \cup T|$, since $\Pr(|S \cup T| = k) = \binom{n}{k}\kappa_k$ for $\Pr(S = A) = \rho_{|A|}$ and $\Pr(T = A) = \nu_{|A|}$. This closure theorem is Walkup's Theorem 4 for two sets [28]: his random sets are conditionally uniform, and their fundamental sequences are ρ and ν. He derives it from his Theorem 2, which proves that the double-binomial convolution $h_k = \sum_i \sum_r \binom{k}{i}\binom{i}{r}\rho_i\nu_{k-r}$, which is κ_k after the substitution $j = k - r$, preserves log-concavity; Proposition 7.3 expresses it in the Newton basis, and the proof in Appendix D is self-contained. Hoggar's theorem [9] is the analogue for the power basis. In terms of the linear map E with $E\binom{x}{j} = x^j$, the Newton coefficients of f are the coefficients of $E(f)$, and $E(fg)$ is Wagner's diamond product of $E(f)$ and $E(g)$ [3, (4.3)], [27, Theorem 2.3]; so Proposition 7.3 states that the diamond product preserves nonnegative coefficient sequences that are log-concave without internal zeros. For the numerator polynomial $W(f)$ of the generating function $\sum_{x\ge 0} f(x)t^x$, whose coefficients expand f in another basis, the pointwise product preserves real zeros of W by Wagner [27], ultra-log-concavity of its coefficients by Brändén, Ferroni and Jochemko [3], and their log-concavity without internal zeros by Liu and Mao [13].

Log-concave ratios on log-concave profiles. Corollary 7.2 extends to every log-concave Newton profile. Besides Proposition 7.3, the proof uses a sign criterion for quadratic forms in log-concave sequences.

Lemma 7.4. *Let $Q(u) = \sum_{0\le i\le j} q_{ij}u_iu_j$ with $q_{0j} \le 0$ for all j and $q_{ij} \ge 0$ for $2 \le i \le j$, and put $N_W(Q) = \sum_{i\le j,\, i+j=W} q_{ij}$. If $N_W(Q) \ge 0$ for every W, then $Q(u) \ge 0$ for every nonnegative sequence $(u_n)_{n\ge 0}$ that is log-concave without internal zeros.*

Proof. Fix W and put $p_i = u_iu_{W-i}$ for $0 \le i \le W/2$ and $p_{-1} = 0$. Log-concavity without internal zeros makes p nondecreasing. With $T_k = \sum_{k\le i\le W/2} q_{i,W-i}$ we get $\sum_{i\le W/2} q_{i,W-i}p_i = \sum_{k\le W/2} T_k(p_k - p_{k-1})$. Here $T_0 = N_W(Q) \ge 0$, $T_1 = N_W(Q) - q_{0W} \ge 0$, and $T_k \ge 0$ for $k \ge 2$. ∎

The weight sums are the coefficients of Q on geometric sequences: $Q(1, r, r^2, \dots) = \sum_W N_W(Q) r^W$. This Abel argument, with the reduction to the full sum when the negative terms come first, is that of Wang and Yeh [29, Lemma 2.1 and Corollary 2.6].

Theorem 7.5 *(Log-concave ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k \geq 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i)) d_i$ and $c_i = \rho(i) d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 6.1 and Corollary 7.2. It also contains the case $t^2 \geq r\tau$ of Theorem 6.2. The exponential ratios $\rho(x) = r(1 + \varepsilon)^x$ are admitted on every log-concave Newton profile.

Certificate map. Steps (i)–(v) of the proof below, and the corresponding reductions in the proof of Theorem 7.6, hold for every N and use no computer assistance. The remaining obligations, and the Lean declarations that prove them for every N (prefix `Szkraw.` omitted), are these.

- Theorem 7.5, step (vi): the edge bounds, *symbolic* for every window size $a \geq 3$ (`eK_nonneg`, `eL1_nonneg`, `eR_nonneg`, `eR2_nonneg`, `eKpp_nonneg`, `eQs_nonneg`, `eKa_nonneg`, `ePsi_nonneg`, `eQhat_nonneg`).
- Theorem 7.5, step (vii): the windows $3 \leq a \leq 5$, *finite*, one exact certificate for each case (`win_MM1_3_2`, …, `win_MM1_5_5`, `win_S20_3`, `win_S20_4`).
- Theorem 7.6: the tail sums of (7.5), (7.6) and (7.4), *symbolic* in ℓ and n (`cubicB_gen`, `cubicB_two`, …, `cubicB_five`, `cubicB_corner`, `cubicE_gen`, `cubicE_three`, …, `cubicE_six`), and $\Phi_3^{\mathrm{loc}} \geq 0$, *symbolic* in a (`Phi3_nonneg`).
- Corollary 7.7: the modified coefficient in (7.5) for $n \leq 5$, *symbolic* in ℓ; Lean proves it within the declarations for (7.5) above, which are stated under $r\sigma \leq \frac{7}{4} t\tau$.

A symbolic certificate has coefficients that are polynomials or rational functions of the parameters and covers the whole range. A worked example after the proof shows how one obligation becomes a polynomial identity checked by Lean. *To check an obligation, see the supplement [21]*: its certificate inventory gives, for each step, the script and the Lean declaration, and its verification procedure gives the command that reruns the script and the Lean build and axiom audit that check the declarations; the exact cross-checks there recompute coefficients at finitely many sizes.

Proof. Setup. Only $\rho_0, \dots, \rho_N$ enter, and truncating (ρ_k) after index N keeps it log-concave without internal zeros, so we may assume it finitely supported. Write $f_n(\ell) = \Delta^n d(\ell)$, $R_i = d_i / f_1(i)$, which is nondecreasing as in the proof of Theorem 6.1, and $P_k(x) = \Delta^k \rho(x) = \sum_h \binom{x}{h} \rho_{k+h}$. As in the proof of Theorem 6.2, $1 - \rho(m+1) \geq \Delta\rho(m) R_m$ for $0 \leq m < N$, and $0 < \rho_0 \leq \rho(i) \leq 1$ for $i \leq N$; ρ is nondecreasing.

For every integer $x \geq 0$, the sequence $(P_k(x))_k$ arises from (ρ_k) by the shifts of Lemma 3.3, so it is log-concave without internal zeros. Since $\rho(x+1)^2 - \rho(x)\rho(x+2) = P_1(x)^2 - P_0(x)P_2(x) \geq 0$, the positive sequence $(\rho(n))_{n\geq 0}$ is log-concave. (The function $x \mapsto \rho(x)$ of a real variable need not be: $\rho(x) = 1 + x/2 + x^2/2$ has $(\log\rho)''(0) > 0$.) Log-concavity in k gives $\rho(y)P_{k+1}(y) \leq P_1(y)P_k(y)$. Combining these facts, for integers $0 \leq x \leq y$ and $k \geq 1$,

$$\rho(x)P_{k-1}(y+1) \leq \rho(x+1)P_{k-1}(y), \qquad \rho(x)P_k(y) \leq \Delta\rho(x)P_{k-1}(y). \tag{7.1}$$

Indeed, log-concavity of $x \mapsto \rho(x)$ gives $\rho(x)\rho(y+1) \leq \rho(x+1)\rho(y)$, equivalently $\rho(x)P_1(y) \leq \Delta\rho(x)\rho(y)$. For the first inequality, $\rho(y)P_{k-1}(y+1) = \rho(y)P_{k-1}(y) + \rho(y)P_k(y) \leq \rho(y+1)P_{k-1}(y)$; multiply by $\rho(x)$ and divide by $\rho(y) > 0$. For the second, $\rho(x)\rho(y)P_k(y) \leq \rho(x)P_1(y)P_{k-1}(y) \leq \Delta\rho(x)\rho(y)P_{k-1}(y)$. By Proposition 7.3, the Newton coefficients of $c = \rho d$ are log-concave without internal zeros, and so is $(\varphi_n(\ell))_n$ for every ℓ. For $\ell + n \leq N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \sum_{j=0}^{n} \binom{n}{j} P_j(\ell+n-j) f_{n-j}, \qquad \psi_n = h_n - v_n - \widehat{w}_n,$$

with $h_n = (1-\rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $\widehat{w}_n = \sum_{j=2}^{n} w_n^{(j)}$, where $w_n^{(j)} = \binom{n}{j}P_j(\ell+n-j)f_{n-j}$. Here φ_n, h_n, v_n, $w_n^{(j)}$ and $\widehat{w}_n$ are nonnegative, while ψ_n need not be.

Moves and detours. Split the moves as in the proof of Theorem 6.2, with W_j now formed from $\widehat{w}_n$. Lemma 3.3 for φ gives $B_0(\ell,a)B_{j-1}(\ell+1,a-2) \geq 4c_{\ell+1}B_j(\ell,a)$ for $j \geq 1$, as in Lemma 3.4. The detours of H and C', with constants 10 and 6, hold verbatim: they used only $\varphi_2(\ell) \geq \rho(\ell+1)f_2(\ell)$, Lemma 3.3 for f, and $0 \leq \rho \leq 1$ nondecreasing.

For V, the term $\rho(\ell+2)f_2$ of φ_2, the first inequality of (7.1) with $k=2$, $x = \ell+1$, $y = \ell+n$, and Lemma 3.3 for f with indices $(2, n-1)$ give

$$\varphi_2(\ell)v_n(\ell+1) \geq \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell).$$

For $\widehat{w}$, expand $\varphi_2(\ell) = \rho(\ell+2)f_2 + 2\Delta\rho(\ell+1)f_1 + P_2(\ell)f_0$, and pair its three terms with the components of $\widehat{w}_n(\ell+1)$:

- $\rho(\ell+2)f_2$ with component $i \leq n$, by the first inequality of (7.1) and indices $(2, n-i)$, gives at least $\binom{n}{i}\binom{n+2}{i}^{-1} c_{\ell+1}w_{n+2}^{(i)}(\ell)$;
- $2\Delta\rho(\ell+1)f_1$ with component $i \leq n$, by the second inequality of (7.1) and indices $(1, n-i)$, gives at least $2\binom{n}{i}\binom{n+2}{i+1}^{-1} c_{\ell+1}w_{n+2}^{(i+1)}(\ell)$;
- $P_2(\ell)f_0$ with component n gives at least $c_{\ell+1}w_{n+2}^{(n+2)}(\ell)$, since $P_2(\ell)P_n(\ell+1) \geq (P_0 + P_1)(\ell)P_{n+2}(\ell)$ by log-concavity in k, and $f_0(\ell+1) = d_{\ell+1}$.

When $n - i = 0$ the index condition is trivial. Collecting by target,

$$\varphi_2(\ell)\widehat{w}_n(\ell+1) \geq c_{\ell+1}\sum_{i=2}^{n+2} \theta_{n,i} w_{n+2}^{(i)}(\ell),$$

where $\theta_{n,2} = \binom{n}{2}\binom{n+2}{2}^{-1}$, $\theta_{n,i} = 1 - \frac{i(i-1)}{(n+1)(n+2)}$ for $3 \le i \le n$, $\theta_{n,n+1} = \frac{2}{n+2}$ and $\theta_{n,n+2} = 1$.

As in the proof of Theorem 6.2, the coefficient quotients are $\frac{j(m-1)(m-2)}{2(m-j-1)}$ for V, and $\theta_{m-2,i}\frac{jm(m-1)}{2(m-j-1)}$ for $\widehat{W}$, on $m \ge 2j+1$, $j \ge 2$. The first is at least 6, by $(m-4)(m-5) \ge 0$ at $j = 2$. The second exceeds 2: it is at least $\frac{j(m-2)(m-3)}{2(m-j-1)} \ge m-2$ for $i = 2$, at least $\frac{j(2m-3)}{m-j-1}$ for $3 \le i \le m-2$, equal to $\frac{j(m-1)}{m-j-1}$ for $i = m-1$, and at least 10 for $i = m$. Hence

$$B_0V'_{j-1} \ge 6c_{\ell+1}V_j \qquad \text{and} \qquad B_0W'_{j-1} \ge 2c_{\ell+1}W_j \qquad (j \ge 2).$$

Resummation. We prove below, for $(\ell, a) \in \Omega$ with $a \ge 3$,

$$A_0^\psi(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^\psi(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1 - 2V_1\big)(\ell,a). \tag{7.2}$$

With the notation of the proof of Theorem 6.2, the inequalities above give $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge 6V_-$, $W_+P \ge 2W_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 \ge 4H_1$. Since $(I - P/\mu)G = (1 - \frac{2}{\mu})G + \frac{2}{\mu}I$, one has $G(H_+ - H_-)G \ge 0$, $GCG \ge 0$, $G(W_+ - W_-)G \ge 0$ and $G(V_+ - V_-)G \ge \frac{2}{3}GV_1G$, so

$$GAG \ge G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \ge 0,$$

and (P) follows as before.

Proof of (7.2): notation. Use the notation of the proof of Theorem 6.1: $u_n = f_n(\ell)$, which is log-concave without internal zeros, y_n, D, X, X', Y, Y', J, H_*. Put also $A = d_{\ell+a}$, $B = d_{\ell+a-1}$, $F = A - B = f_1(\ell+a-1)$, $X'' = B - d_\ell$ and $P_k = P_k(\ell)$. Since $c_{\ell+n} = d_{\ell+n}\sum_k \binom{n}{k}P_k$ for $n \le a$, the following are linear in P:

- $B_0(\ell,a) = \sum_k \beta_k P_k$ and $B_0(\ell+1,a-1) = \sum_k \beta'_k P_k$, where
$$\begin{aligned} &\beta_0 = Y, \quad \beta_1 = a(A-D), \quad \beta_k = \tbinom{a}{k}A \ (k \ge 2),\\ &\beta'_0 = Y', \quad \beta'_1 = aA - 2(a-1)d_{\ell+2} + (a-2)D,\\ &\beta'_2 = \tbinom{a}{2}A - (a-1)d_{\ell+2}, \quad \beta'_k = \tbinom{a}{k}A \ (k \ge 3); \end{aligned}$$
- $c_{\ell+a} - c_\ell = \sum_k \chi_k P_k$ and $c_{\ell+a-1} - c_{\ell+1} = \sum_k \chi'_k P_k$;
- $c_{\ell+a} - c_{\ell+a-1} = \sum_k \delta_k P_k$ with $\delta_k = \binom{a}{k}F + \binom{a-1}{k-1}B$;
- $H_1 = H_* - \sum_k \zeta_k P_k$ with $\zeta_k = \sum_{n\ge3}(n-2)\binom{n}{k}y_n$, and $V_1 = \sum_k \nu_k P_k$ with $\nu_k = \sum_{n\ge3}(n-2)n\binom{n-1}{k-1}\binom{a}{n}u_{n-1}$;
- $c_{\ell+1} = D(P_0 + P_1)$.

As $A_0^\psi(\ell,a) = X - (c_{\ell+a} - c_\ell)$ and $A_0^\psi(\ell+1,a-2) = X' - (c_{\ell+a-1} - c_{\ell+1})$, the difference of the two sides of (7.2) is $\mathrm{Key}(P) = \mathrm{Lin}(P) - \mathrm{Quad}(P)$. Here $\mathrm{Lin}(P) = \sum_k l_k P_k$ with $l_k = X\beta'_k + \beta_k X' - 4[k \le 1]DH_*$, and Quad is a quadratic form. We use repeatedly:

$$AX'' \ge (a-1)d_\ell F, \qquad AX' \ge (a-2)DF, \qquad u_0F \le u_1B, \qquad FD \le u_1A. \tag{7.3}$$

The first two hold because $A/F \geq R_{\ell+a-1} \geq R_{\ell+i}$ gives $(A/F)f_1(\ell+i) \geq d_{\ell+i}$ for $i \leq a-1$; summing over $0 \leq i \leq a-2$, respectively $1 \leq i \leq a-2$, gives them. The last two follow from $u_0u_{m+1} \leq u_1u_m$, as $F = \sum_m \binom{a-1}{m}u_{m+1}$, $B = \sum_m \binom{a-1}{m}u_m$ and $D = u_0 + u_1$.

(i) Reduction to a quadratic form in P. By monotonicity of b at $\ell+a-1$, $\Delta c := c_{\ell+a} - c_{\ell+a-1} \leq F$. With $\Phi(P) := \Delta c \cdot \mathrm{Lin}(P) - F\,\mathrm{Quad}(P)$ one has $\mathrm{Key} = (1 - \Delta c/F)\mathrm{Lin} + \Phi/F$, so it suffices that every $l_k \geq 0$ and $\Phi \geq 0$. With $\chi_k = \binom{a}{k}A - [k=0]d_\ell$ and $\chi'_k = \binom{a-1}{k}B - \binom{1}{k}D$, so that $E^1_m = X\delta_m - F\chi_m$, $E^2_m = X'\delta_m - F\chi'_m$ and $E^3_m = 2F\nu_m - 4(H_*\delta_m - F\zeta_m)$, expansion gives $\Phi(P) = \sum_{k,m} M_{k,m}P_kP_m$ with

$$M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m + [k \leq 1]D\,E^3_m,$$

and explicitly $E^1_0 = E^2_0 = E^3_0 = 0$ and, for $m \geq 1$,

$$E^1_m = \tbinom{a-1}{m-1}AX'' - \tbinom{a-1}{m}d_\ell F, \qquad E^2_m = \tbinom{a-1}{m-1}AX' - \Big(\tbinom{a-1}{m} - [m=1]\Big)DF,$$
$$E^3_m = 2F\nu_m - 4\tbinom{a-1}{m-1}BH_* - 4FZ_m, \qquad Z_m = \sum_{n\geq 3}(n-2)\Big(\tbinom{a}{m} - \tbinom{n}{m}\Big)y_n.$$

Since (P_k) is log-concave without internal zeros, the Abel summation in the proof of Lemma 7.4, applied to every anti-diagonal of Φ, shows $\Phi \geq 0$ if every central sum $S_{\sigma,i} = \sum_{k=i}^{\sigma-i} M_{k,\sigma-k}$, $0 \leq i \leq \sigma/2$, is nonnegative. As $M_{k,0} = 0$, for $\sigma \geq 3$

$$S_{\sigma,0} = S_{\sigma,1} + M_{0,\sigma} \qquad \text{and} \qquad S_{\sigma,1} = S_{\sigma,2} + M_{1,\sigma-1} + M_{\sigma-1,1}.$$

Since $\binom{a-1}{m} = \frac{a-m}{m}\binom{a-1}{m-1}$ and $\frac{a-m}{m(a-1)} \leq \frac{a-m}{m(a-2)} \leq \frac{1}{2}$ for $m \geq 2$, (7.3) gives, for $m \geq 2$,

$$E^1_m \geq \tfrac{1}{2}\tbinom{a-1}{m-1}AX'' \geq 0 \qquad \text{and} \qquad E^2_m \geq \tfrac{1}{2}\tbinom{a-1}{m-1}AX' \geq 0.$$

Also every $\beta_k \geq 0$ and $\beta'_k \geq 0$; for β'_2 use $\binom{a}{2}A \geq (a-1)d_{\ell+2}$. Finally $E^1_1 = AX'' - (a-1)d_\ell F \geq 0$ and $E^2_1 = AX' - (a-2)DF \geq 0$.

(ii) The central sums with $i \geq 2$, and the coefficients l_k. For $k, m \geq 2$, $M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m \geq 0$, so $S_{\sigma,i} \geq 0$ for $i \geq 2$. Next, $l_0 = K \geq 0$ by the proof of Theorem 6.1, and $l_k \geq 0$ for $k \geq 2$ termwise. By convexity, $A - D \geq (a-1)f_1(\ell+1) = (a-1)(d_{\ell+2} - D)$, so $\beta'_1 \geq (a-2)(A-D)$ and

$$l_1 \geq L := (A-D)\big((a-2)X + aX'\big) - 4DH_*.$$

(iii) The central sums with $\sigma \leq 2$. Take $P = (P_0, P_1, 0, \ldots)$ with $P_0 \geq 0$ and $P_1 > 0$, and put $\lambda = F/\Delta c(P)$. Then $\mathrm{Key}(\lambda P) = (\lambda/\Delta c)\Phi(P)$, and λP is the linear ratio $\tilde\rho(\ell + x) = \lambda(P_0 + P_1 x)$. Its $\tilde b = (1-\tilde\rho)d$ satisfies $\Delta\tilde b(\ell+a-1) = 0$. So $\alpha := 1 - \tilde\rho(\ell+a) = tR_{\ell+a-1} \geq tu_0/u_1$ with $t = \lambda P_1$. For such α and t, the proof of Theorem 6.1 shows (6.2). Its two weights A^ψ_0 are nonnegative, since $X \geq au_1$ and $X' \geq (a-2)u_1$, and $B_0(\ell, a) \geq \tilde\rho(\ell+1)Y$, $B_0(\ell+1, a-1) \geq \tilde\rho(\ell+1)Y'$, since $\beta_1 \geq Y$ and $\beta'_1 \geq Y'$. Hence $\mathrm{Key}(\lambda P) \geq 0$, that is, $\Phi(P) = P_1(M_{0,1}P_0 + M_{1,1}P_1) \geq 0$. This gives $S_{1,0} = M_{0,1} \geq 0$ and $S_{2,1} = M_{1,1} \geq 0$. The sum $S_{2,0} = M_{0,2} + M_{1,1}$ is treated in (v).

(iii′) $M_{1,m}+M_{m,1}\ge 0$ *for* $m\ge 2$ *and* $a\ge 6$. Put $c'_m=\binom{a-1}{m-1}$. Then

$$M_{1,m}+M_{m,1}=\big[\beta_1E^2_m-4c'_mDBH_*\big]+\big[\beta'_1E^1_m-DF(4Z_m-2\nu_m)\big]+\beta'_mE^1_1+\beta_mE^2_1,$$

and the last two terms are nonnegative.

- First bracket: $E^2_m\ge\frac12c'_mAX'$ and $A\ge B$, so it is nonnegative if $R:=a(A-D)X'-8DH_*\ge 0$.
- Second bracket: $Z_m\le\binom{a}{m}H_*\le\frac{a}{2}c'_mH_*$, $\nu_m\ge 0$, $F\le A$, $E^1_m\ge\frac12c'_mAX''$ and $\beta'_1\ge(a-2)(A-D)$. So it is nonnegative if $R_2:=(a-2)(A-D)X''-4aDH_*\ge 0$.

(iv) $M_{0,s}\ge 0$ *for* $s\ge 3$. Put $c'=\binom{a-1}{s-1}$ and $c''=\binom{a-1}{s}$. Then

$$M_{0,s}=Y'E^1_s+YE^2_s+DE^3_s=c'BK''+FG_s,$$

where

$$K''=Y'X''+YX'-4DH_*,\quad G_s=c'(Y'X''+YX')-c''(d_\ell Y'+DY)+D(2\nu_s-4Z_s).$$

Note $u_1=d_{\ell+1}-d_\ell>0$. If $K''\ge 0$, then $u_1B\ge u_0F$ gives $u_1M_{0,s}\ge F(c'u_0K''+u_1G_s)$, and

$$c'u_0K''+u_1G_s=D\,Q_s+c''u_1^2Y',$$
$$Q_s=c'(Y'X''+YX')-4c'u_0H_*+u_1(2\nu_s-4Z_s)-c''u_1(Y+Y').$$

So $M_{0,s}\ge 0$ once $K''\ge 0$ and $Q_s\ge 0$.

(v) $S_{2,0}\ge 0$ *for* $a\ge 5$. With $c''=\binom{a-1}{2}$, the definitions give the identity

$$S_{2,0}=\beta'_1E^1_1+\big[\beta_1E^2_1-c''F(d_\ell Y'+DY)\big]+B\,K_a+F\,W,$$

where $K_a=(a-1)K''-4DH_*$ and $W=(a-1)(Y'X''+YX')+D(2\nu_1+2\nu_2-4Z_1-4Z_2)$.

- Since $\beta_1=a(X'+F)$, $E^2_1=AX'-(a-2)DF$, $d_\ell\le D$ and $FD\le u_1A$, the bracket is at least $A\Psi$, where $\Psi=a(A-D)J-c''u_1(Y+Y')$. Recall $J=X'-(a-2)u_1$ and $X'+F=A-D$.
- If $K_a\ge 0$, then $u_1(BK_a+FW)\ge F(u_0K_a+u_1W)=FD\widehat{Q}$, where $\widehat{Q}=(a-1)(Y'X''+YX')-4au_0H_*+u_1(2\nu_1+2\nu_2-4Z_1-4Z_2)$.

Hence $u_1S_{2,0}\ge u_1\beta'_1E^1_1+u_1A\Psi+FD\widehat{Q}\ge 0$ once K_a, Ψ and $\widehat{Q}$ are nonnegative.

(vi) Edge bounds. Steps (ii)–(v) reduce (7.2) to the nonnegativity of the quadratic forms K, L, R, R_2, K'', Q_s $(3\le s\le a)$, K_a, Ψ and $\widehat{Q}$ in the sequence u. For Q_s, first remove the dependence on s: by $\nu_s\ge 0$, $Z_s\le\binom{a}{s}H_*$, $\binom{a}{s}\le\frac{a}{3}\binom{a-1}{s-1}$ and $\binom{a-1}{s}\le\frac{a-3}{3}\binom{a-1}{s-1}$ for $s\ge 3$, one has $Q_s\ge\binom{a-1}{s-1}Q^*$ with

$$Q^*=Y'X''+YX'-4u_0H_*-\tfrac{4a}{3}u_1H_*-\tfrac{a-3}{3}u_1(Y+Y').$$

Each of these forms is a sum of products xy of linear forms $x = \sum_{i\ge1} x_i u_i$ and $y = \sum_{j\ge1} y_j u_j$ with nonnegative coefficients, plus u_1 times a linear form, minus u_0 times a linear form with nonnegative coefficients. Since u is log-concave without internal zeros, $u_i u_j \ge u_1 u_{i+j-1}$ for $i, j \ge 1$ and $u_0 u_{n+1} \le u_1 u_n$. Keeping in xy only the pairs with $i \le 2$ or $j \le 2$ gives

$$xy \ge u_1 \sum_n e_n(x,y) u_n, \qquad e_n(x,y) = x_1 y_n + x_2 y_{n-1} + [n \ge 3]\, y_1 x_n + [n \ge 4]\, y_2 x_{n-1},$$

and $u_0 \sum_{n\ge1} h_n u_n \le u_1 \sum_{n\ge0} h_{n+1} u_n$. So each form is at least $u_1 \sum_n e_n u_n$ with explicit coefficients e_n, and it suffices that every $e_n \ge 0$.

- For $4 \le n \le a$ write $n = 4 + p$ and $a = n + k$. Then $e_n / \binom{a}{n}$ is a rational function of (p, k) whose numerator has nonnegative coefficients and whose denominator is strictly positive for $p, k \ge 0$.
- For $n \le 3$, e_n is a polynomial in a whose coefficients are nonnegative after $a \mapsto a + a_0$, with $a_0 = 6$ for R_2, $a_0 = 5$ for K_a and $a_0 = 3$ otherwise.
- For $n \ge a + 1$ only nonnegative products remain.

These certificates are formalized in Lean, and `runs/lean_plan/edge_exact.py` checks every e_n in exact arithmetic for $a \le 60$.

(vii) Small windows. Steps (iii′) and (v) need $a \ge 6$ and $a \ge 5$. For $3 \le a \le 5$, each sum $M_{1,m} + M_{m,1}$ $(2 \le m \le a)$ and, for $a \le 4$, $S_{2,0}$ is a cubic form in u that is an exact nonnegative rational combination of the products $u_k(u_{i+1}u_{j-1} - u_i u_j)$ $(i + 1 \le j - 1)$ and of monomials (`runs/lean_plan/window_cert.py` in [18]). This proves (7.2) and the theorem. ∎

A worked certificate. The smallest obligation shows how a step of the proof becomes a polynomial identity checked by Lean. In step (vii) with $a = 3$, the central sum $S_{2,0} = M_{0,2} + M_{1,1}$ must be nonnegative. Substituting the definitions of step (i), with $u_n = f_n(\ell)$, gives a cubic form in $u_0, \dots, u_3$ with integer coefficients,

$$\begin{aligned} S_{2,0} = {} & 24u_1^3 + 135u_1^2u_2 + 189u_1u_2^2 + 66u_2^3 + 21u_1^2u_3 + 98u_1u_2u_3 + 52u_2^2u_3 \\ & + 11u_1u_3^2 + 10u_2u_3^2 + 12u_0u_1u_2 + 12u_0u_2^2 - 42u_0u_1u_3 - 29u_0u_2u_3 \\ & - 11u_0u_3^2 - 12u_0^2u_3, \end{aligned}$$

and the obligation is $S_{2,0} \ge 0$ for every nonnegative sequence u that is log-concave without internal zeros. For such u the products $u_k(u_{i+1}u_{j-1} - u_i u_j)$ with $i + 1 \le j - 1$ are nonnegative, and so are all monomials. The script `runs/lean_plan/window_cert.py` solves an exact linear program for nonnegative rational multipliers of these generators and returns

$$\begin{aligned} S_{2,0} = {} & 36u_0(u_1u_2 - u_0u_3) + 42u_0(u_2^2 - u_1u_3) + 24u_1(u_1^2 - u_0u_2) + 189u_1(u_2^2 - u_1u_3) \\ & + 135u_2(u_1^2 - u_0u_2) + 29u_3(u_1^2 - u_0u_2) + 98u_3(u_1u_2 - u_0u_3) \\ & + 24u_0^2u_3 + 105u_0u_2^2 + 87u_0u_3^2 + 181u_1^2u_3 + 11u_1u_3^2 + 66u_2^3 + 52u_2^2u_3 + 10u_2u_3^2, \end{aligned}$$

an identity of polynomials, which the script checks by expansion. The Lean statement `Szkraw.win_S20_3` asserts $S_{2,0} \geq 0$ for every such sequence. Its proof records the fifteen generators above as nonnegative facts, expands $S_{2,0}$ from the Lean definitions of the $M_{k,m}$, and closes the goal by linear arithmetic, which the Lean kernel checks. The symbolic certificates of step (vi) use the transfers $u_i u_j \geq u_1 u_{i+j-1}$ and $u_0 u_{n+1} \leq u_1 u_n$ in place of these generators, and their coefficients e_n are rational functions of a and n whose numerators have nonnegative coefficients, so one certificate covers every window size.

For cubic ratios, log-concavity of the Newton coefficients can be replaced by a single inequality. We state the bound $r\sigma \leq t\tau$ first because it follows from log-concavity of (r,t,τ,σ) and its written proof needs only tail-sum certificates; the stronger bound $r\sigma \leq \frac{7}{4}t\tau$ of Corollary 7.7 needs further certificates for small n. The Lean proof establishes the corollary directly and derives the theorem from it.

Theorem 7.6 *(Cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \geq 0$ and $r\sigma \leq t\tau$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

If (r,t,τ,σ) is log-concave without internal zeros, then $r\sigma \leq t\tau$: for $t\tau > 0$ this follows from $t^2\tau^2 \geq r\tau \cdot t\sigma$, and otherwise $\sigma = 0$. Theorem 7.6 also admits cubic ratios with $t^2 < r\tau$ or $\tau^2 < t\sigma$, which Theorem 7.5 does not cover. The failing cubic examples of Appendix C violate $r\sigma \leq t\tau$. The proof is computer-assisted: the inequalities (7.5), (7.6) and (7.4) and the inequality $\Phi_3^{\mathrm{loc}} \geq 0$ below are reduced to families of explicit polynomials with nonnegative coefficients.

Proof. Setup. As in the proof of Theorem 6.2, $R_i = d_i/f_1(i)$ is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m) R_m$ for $0 \leq m < N$, and ρ is nondecreasing with $0 < \rho(i) \leq 1$ for $i \leq N$; here $\Delta\rho(x) = t + \tau x + \sigma\binom{x}{2}$ and $\Delta^2\rho(x) = \tau + \sigma x$. For $\ell + n \leq N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell+n) f_n + n\Delta\rho(\ell+n-1) f_{n-1} + \tbinom{n}{2}\Delta^2\rho(\ell+n-2) f_{n-2} + \tbinom{n}{3}\sigma f_{n-3}.$$

Since $\binom{\ell+n-1}{2} = \binom{\ell}{2} + \ell(n-1) + \binom{n-1}{2}$ and $n\binom{n-1}{2} = \binom{n}{2}(n-2) = 3\binom{n}{3}$, this gives $\psi_n = h_n - v_n - e_n$ with $h_n = (1-\rho(\ell+n)) f_n$,

$$v_n = n\Gamma_n(\ell) f_{n-1}, \qquad \Gamma_n(\ell) = t + \tau(\ell+n-1) + \sigma\Big(\tbinom{\ell}{2} + \ell(n-1)\Big),$$
$$e_n = \tbinom{n}{2}(\tau + \sigma\ell) f_{n-2} + \sigma\tbinom{n}{3}\big(3f_{n-1} + 3f_{n-2} + f_{n-3}\big),$$

all nonnegative. Write $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$; then (u_k) is log-concave, and $u_0 \leq \ell u_1$ by convexity.

Moves. Split the moves as in the proof of Theorem 6.2, with V_j formed from v_n and with a third family $E_j = W_{2j+1,j}$ formed from e_n in place of W_j.

Detours of H, V and C'. The detours of H and C' hold verbatim, with constants 10 and 6. For V,

$$(n+1)\Gamma_n(\ell+1) - n\Gamma_{n+2}(\ell) = t + \tau\ell + \sigma\Big(\tbinom{\ell}{2} + n^2 - 1\Big) \geq 0,$$

so Lemma 3.3 with indices $(2, n-1)$ gives $(n+1)(n+2)\varphi_2 v'_n \geq n^2 c_{\ell+1} v_{n+2}$ for $n \geq 3$, and $B_0 V'_{j-1} \geq \frac{9}{2} c_{\ell+1} V_j$ for $j \geq 2$ as before. (With the full σ-part $n\sigma\binom{\ell+n-1}{2} f_{n-1}$ in v_n, the corresponding difference is negative for $\ell < n+1$; without any σ-part in v_n, the reduction of the key inequality below fails for large ℓ.)

Detour of E. We show, for $n \geq 3$ and $\ell + n + 2 \leq N$,

$$(n+1)(n+2)\Big(\varphi_2(\ell) + \frac{n}{3}\varphi_3(\ell)\Big) e_n(\ell+1) \;\geq\; 3(n-1)\, c_{\ell+1} e_{n+2}(\ell). \tag{7.4}$$

Granting this, $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell) + \binom{a}{3}\varphi_3(\ell)$, $\binom{a}{3}\binom{a-2}{n} = \frac{a-2}{3}\binom{a}{2}\binom{a-2}{n}$ with $a - 2 \geq n$ whenever $\binom{a}{n+2} \neq 0$, and $\binom{a}{2}\binom{a-2}{n} = \binom{a}{n+2}\binom{n+2}{2}$ give, for $j \geq 2$,

$$B_0(\ell, a) E_{j-1}(\ell+1, a-2) \;\geq\; \tfrac{3}{2} c_{\ell+1} \sum_{m \geq 2j+1} (m-3)\tbinom{m-j-2}{j-1}\tbinom{a}{m} e_m(\ell) \;\geq\; 3 c_{\ell+1} E_j(\ell, a),$$

because $(m-3)\binom{m-j-2}{j-1} - 2\binom{m-j-1}{j} = \binom{m-j-2}{j-1}\frac{(j-2)(m-1)}{j} \geq 0$.

Detour of B. We show, for $n \geq 2$, $\ell + n + 2 \leq N$ and $(\ell, n) \neq (0, 2)$,

$$(n+1)(n+2)\, \varphi_2(\ell)\varphi_n(\ell+1) \;\geq\; 8(n-1)\, c_{\ell+1}\varphi_{n+2}(\ell), \tag{7.5}$$

and, if $N \geq 4$,

$$\big(12\varphi_2(0) + 8\varphi_3(0)\big)\varphi_2(1) \;\geq\; 8 c_1 \varphi_4(0). \tag{7.6}$$

Granting these, the argument after (6.3) gives $B_0(\ell, a) B_{j-1}(\ell+1, a-2) \geq 4 c_{\ell+1} B_j(\ell, a)$ for $j \geq 1$. The only exception is the term $m = 4$ for $\ell = 0$, $j = 1$. There $B_0(0, a) \geq \binom{a}{2}\varphi_2(0) + \binom{a}{3}\varphi_3(0)$, $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$ and $\binom{a}{3}\binom{a-2}{2} = 2(a-2)\binom{a}{4} \geq 4\binom{a}{4}$ reduce the term to (7.6). Inequality (7.5) fails at $(\ell, n) = (0, 2)$ in general.

Proof of (7.5), (7.6) and (7.4). In each, the difference of the two sides is a quadratic form $\sum \kappa_{\beta\gamma}\rho_\beta\rho_\gamma$ in $(\rho_0, \rho_1, \rho_2, \rho_3) = (r, t, \tau, \sigma)$. Its coefficients are bilinear forms in the u_k, with coefficients polynomial in ℓ and n.

- Since $t\tau\kappa_{t\tau} + r\sigma\kappa_{r\sigma} = (t\tau - r\sigma)\kappa_{t\tau} + r\sigma(\kappa_{t\tau} + \kappa_{r\sigma})$ and $r\sigma \leq t\tau$, the form is nonnegative if $\kappa_{t\tau}$, $\kappa_{t\tau} + \kappa_{r\sigma}$ and all coefficients other than $\kappa_{r\sigma}$ are nonnegative as forms in u.

- For (7.5) the coefficients κ_{rr}, $\kappa_{r\tau}$, $\kappa_{\tau\tau}$ need not be checked: the form with $t = \sigma = 0$ is (6.3) for the ratio $r + \tau\binom{x}{2}$. The same holds for (7.6), whose left side exceeds that of (7.5) at $(0, 2)$ by $8\varphi_3(0)\varphi_2(1) \geq 0$.

- Every remaining coefficient is a sum $\sum c_{ij} u_i u_j$ over $i \leq j$. Group its terms by $i + j = m$. Within a group, $u_i u_{m-i}$ is nondecreasing in i by log-concavity, so with the tail sums $T_k = \sum_{i \geq k} c_{i,m-i}$ and the smallest index k_0 of the group,

$$\sum_i c_{i,m-i} u_i u_{m-i} \;=\; T_{k_0} u_{k_0} u_{m-k_0} + \sum_{k > k_0} T_k\,(u_k u_{m-k} - u_{k-1} u_{m-k+1}),$$

  and the coefficient is nonnegative if all $T_k \geq 0$.

- The tail sums are polynomials in (ℓ, n). For (7.5) with $n \geq 6$ and for (7.4) with $n \geq 7$ they have nonnegative coefficients in $(\ell, n-6)$, respectively $(\ell, n-7)$. For each n in $3 \leq n \leq 5$ ((7.5)) and $3 \leq n \leq 6$ ((7.4)) they are polynomials in ℓ with nonnegative coefficients. The same holds for (7.5) with $n = 2$ and $\ell \geq 1$ after $\ell \mapsto \ell + 1$, and for (7.6), where $u_0 = 0$.

The scripts `runs/lean_plan/gen_cubicB.py` and `runs/lean_plan/gen_cubicE.py` check every tail-coefficient sign in exact rational arithmetic and generate the Lean certificates; Lean checks the certificate identities.

Key inequality. We show, for $(\ell, a) \in \Omega$ with $a \geq 3$,

$$A_0^{\psi}(\ell, a) B_0(\ell+1, a-1) + B_0(\ell, a) A_0^{\psi}(\ell+1, a-2) \;\geq\; c_{\ell+1}\big(4H_1 - 2V_1 - E_1\big)(\ell, a). \quad (7.7)$$

Use the notation of the proof of Theorem 6.2, and put $P_1 = \Delta\rho(\ell)$ and $P_2 = \Delta^2\rho(\ell) = \tau + \sigma\ell$, so that $\rho(\ell + k) = \rho(\ell) + kP_1 + \binom{k}{2}P_2 + \binom{k}{3}\sigma$. Put $\alpha = 1 - \rho(\ell + a)$, $s_q = P_1 + (a-1)P_2$ and $s = \Delta\rho(\ell + a - 1) = s_q + \binom{a-1}{2}\sigma$. Then $\alpha \geq s u_0/u_1$ and $s_q \geq (a-1)P_2$. Backward differences give

$$\begin{aligned} A_0^{\psi}(\ell, a) \;&=\; \alpha X - s_q a u_0 + P_2\tbinom{a}{2}u_0 - \sigma\tbinom{a}{3}u_0, \\ A_0^{\psi}(\ell+1, a-2) \;&=\; (\alpha + s)X' - s_q(a-2)D + P_2\tbinom{a-1}{2}D - \sigma\tbinom{a-1}{3}D, \end{aligned}$$

and, reindexing as before,

$$4H_1 - 2V_1 - E_1 \;=\; 4\alpha H_* + 2s_q T + 4P_2 Z - P_2 \sum_{n \geq 1} n\tbinom{a-n}{2} y_n + \sigma(4T_3 - E_3),$$

where $T_3 = \sum_{n \geq 3}(n-2)\big(\binom{a}{3} - \binom{n}{3}\big)y_n$ and $E_3 = \sum_{n \geq 3}(n-2)\binom{n}{3}\binom{a}{n}(3u_{n-1} + 3u_{n-2} + u_{n-3})$. With $B_0(\ell, a) \geq pY$, $B_0(\ell+1, a-1) \geq pY'$ and $c_{\ell+1} = pD$, it suffices that $R = A_0^{\psi}(\ell, a)Y' + YA_0^{\psi}(\ell+1, a-2) - D(4H_1 - 2V_1 - E_1) \geq 0$. R is affine in α, with coefficient $XY' + YX' - 4DH_*$. Its part without σ is the expression bounded in the proof of Theorem 6.2, with s_q and P_2 in place of s and τ; that bound holds for every value α' of α with $\alpha' u_1 \geq s_q u_0$. Apply it with $\alpha' = \alpha - \binom{a-1}{2}\sigma z$, where $z = u_0/u_1$. This gives $R \geq \sigma\Phi_3^{\mathrm{loc}}$ with

$$\Phi_3^{\mathrm{loc}} \;=\; \tbinom{a-1}{2} z\,(XY' + YX' - 4DH_*) - \tbinom{a}{3}u_0 Y' + Y\Big(\tbinom{a-1}{2}X' - \tbinom{a-1}{3}D\Big) - D(4T_3 - E_3).$$

$\Phi_3^{\mathrm{loc}} \geq 0$. Write $E_3 = \binom{a}{3}u_0 + E_3^0$ and $H_*^+ = \sum_{n \geq 2}(n-1)\binom{a}{n+1}u_n$, so that $u_0 H_* \leq u_1 H_*^+$. Bounding the only negative term quadratic in u_0, $-4\binom{a-1}{2}u_0^2 H_*$, this way and dropping $\binom{a}{3}u_0^2 u_1 \geq 0$ gives $u_1\Phi_3^{\mathrm{loc}} \geq u_1 A + u_0 B$ with

$$\begin{aligned} A \;&=\; \tbinom{a-1}{2}YX' - u_1\Big(\tbinom{a-1}{3}Y + 4T_3 - E_3^0\Big), \\ B \;&=\; \tbinom{a-1}{2}\big(XY' + YX' - 4u_1 H_* - 4u_1 H_*^+\big) - u_1\Big(\tbinom{a}{3}Y' + \tbinom{a-1}{3}Y + 4T_3 - E_3^0 - \tbinom{a}{3}u_1\Big). \end{aligned}$$

These are forms of the kind treated in step (vi) of the proof of Theorem 7.5: the edge bounds give $A \geq u_1 \sum_n e_n^A u_n$ and $B \geq u_1 \sum_n e_n^B u_n$. Their coefficients are nonnegative by the same

three kinds of certificates: for $4 \le n \le a$ the numerators in (p,k) have nonnegative coefficients and the denominators are strictly positive for $p,k \ge 0$, for $n \le 3$ the polynomials in a have nonnegative coefficients after $a \mapsto a+3$, and for $n \ge a+1$ only nonnegative terms remain (`runs/lean_plan/phi3_generic.py` and `runs/lean_plan/phi3_edge.py` in [18]). Hence $R \ge 0$.

Resummation. Proceed as in the proof of Theorem 6.2, with E in place of W. The inequalities $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge \frac{9}{2}V_-$, $E_+P \ge 3E_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 + E_1 \ge 4H_1$ give $GAG \ge \frac{1}{2}(GA_0 + A_0G) + \frac{1}{18}GV_1G + \frac{1}{12}GE_1G \ge 0$, and (P) follows. ∎

Corollary 7.7 *(Extended cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \ge 0$ and $r\sigma \le \frac{7}{4}t\tau$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

Proof. The proof of Theorem 7.6 uses $r\sigma \le t\tau$ only in (7.5), (7.6) and (7.4), and there only through $\kappa_{t\tau} + \kappa_{r\sigma} \ge 0$; the other coefficients, the key inequality (7.7) and the resummation do not involve the condition. Since

$$t\tau\,\kappa_{t\tau} + r\sigma\,\kappa_{r\sigma} = \big(t\tau - \tfrac{4}{7}r\sigma\big)\kappa_{t\tau} + \tfrac{4}{7}r\sigma\big(\kappa_{t\tau} + \tfrac{7}{4}\kappa_{r\sigma}\big),$$

under $r\sigma \le \frac{7}{4}t\tau$ it suffices that $\kappa_{t\tau} \ge 0$ and $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma} \ge 0$ as forms in u. For (7.4), for (7.5) with $n \ge 6$ and for (7.6), where $u_0 = 0$, the tail sums of the modified coefficient $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma}$ stay nonnegative. For (7.5) with $n \le 5$ the transfers alone do not suffice. There the modified coefficient is certified after the substitution $u_0 = zu_1$, $0 \le z \le \min(\ell, 1/q)$ with $q = u_2/u_1$, in the regimes $z = 0$, $z = \ell$ and $z = 1/q$ (`runs/cubic/bdetour_u.py` in [18]; `runs/cubic/edetour_u.py` certifies (7.4) independently in the same way). The relevant expressions are affine in z, so these boundary regimes cover the admissible interval for $q > 0$. When $q = 0$, all u_k with $k \ge 2$ vanish, and the inequality follows directly. ∎

The Lean proof, `Szkraw.cubic_ratio74_matrix`, proves Corollary 7.7 directly and derives Theorem 7.6 from it. For (7.5) with $n \le 5$ it does not substitute $u_0 = zu_1$: it writes the modified coefficient, times a positive polynomial in ℓ, as a combination of transfers, products $u_a u_b$ and the terms $u_j(\ell u_1 - u_0) \ge 0$, with multipliers that are polynomials in ℓ with nonnegative coefficients after a shift of ℓ, plus exact rational certificates at the few smaller values of ℓ (`runs/lean_plan/cert74.py` and `runs/lean_plan/gen_cubicB.py` in [18]; see the supplement [21]). Exact diagnostics for both routes are listed in the documentation [19].

Remark 7.8. The factor $\frac{7}{4}$ is a limit of the certificates, not of property (P). On geometric profiles at $(\ell, n) = (0, 3)$ one has $\kappa_{r\sigma} = -20q(4q^3 + 11q^2 + 9q + 1)$ and $\kappa_{t\tau} = 20(q+1)(8q^3 + 33q^2 + 30q + 6)$ in (7.5), so $\kappa_{t\tau}/|\kappa_{r\sigma}| \to 2$ as $q \to \infty$. This family therefore obstructs the coefficientwise test for $r\sigma \le u\,t\tau$ with $u > 2$; it is not a counterexample to property (P). With $u = \frac{9}{5}$ the certificate of `runs/cubic/bdetour_u.py` fails. Whether $r\sigma \le u\,t\tau$ suffices for some $u > \frac{7}{4}$ is open (Section 10).

8. Summary of the criteria

Table 3 lists the criteria of Theorem A and of Sections 5–7 with their hypotheses and the earlier criterion that each extends, and Table 4 gives for each an array or class that it covers and the criteria it extends do not.

Table 3: The criteria of Theorem A and of Sections 5–7 and the earlier criterion that each extends. Each array satisfies (2.1), with profile $d_i = b_i + c_i$ and ratio $\rho(i) = c_i/d_i$. "u LC" means that the Newton coefficients of u are nonnegative and log-concave without internal zeros, and "$b\uparrow$" that b is nondecreasing.

Result	Hypotheses	Extends
Thm. 3.1 (Theorem A)	d LC; $\rho \equiv r \in (0,1]$	—
Thm. 5.1, Cor. 5.2	c LC; Newton coefficients of b are $w_k\kappa_k$ with $w_1 \geq w_2 \geq \cdots \geq 0$	Thm. 3.1 (w constant)
Prop. 5.3	b, c quadratic in i; $\kappa_2 \geq 0$, $\kappa_2 + \beta_2 \geq 0$; $b\uparrow$	Cor. 5.2 on quadratic arrays
Thm. 6.1	d LC; $\rho(i) = r + \kappa\frac{i}{N}$, $\kappa \geq 0$; $b\uparrow$	Prop. 5.3 (linear d)
Thm. 6.2	d LC; $\rho = r + tx + \tau\binom{x}{2}$, $t,\tau \geq 0$; $b\uparrow$	Thm. 6.1 ($\tau = 0$)
Thm. 7.1, Cor. 7.2	$d = \nu_1 i + \nu_2\binom{i}{2}$; c LC (Cor.: ρ LC, any degree); $b\uparrow$	Prop. 5.3 on these profiles
Thm. 7.5 (Theorem B)	d LC; ρ LC, $\rho(0) > 0$; $b\uparrow$	Thm. 6.1, Cor. 7.2, Thm. 6.2 for $t^2 \geq r\tau$
Thm. 7.6, Cor. 7.7	d LC; $\rho = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$, $t,\tau,\sigma \geq 0$, $r\sigma \leq t\tau$ (Cor.: $r\sigma \leq \frac{7}{4}t\tau$); $b\uparrow$	Thm. 6.2 ($\sigma = 0$), Thm. 7.5 for cubic ρ

Table 4: For each criterion of Table 3, an array or class that it covers and the criteria it extends do not.

Result	Newly covered
Thm. 3.1	$K_{N,r}$ and $G_{N,q,r}$
Thm. 5.1, Cor. 5.2	c convex quadratic, $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$ with $w_1 > w_2 \geq 0$
Prop. 5.3	$d_i = \frac{i}{N}$, $\rho(i) = r + \kappa\frac{i}{N}$ with $0 < \kappa \leq \frac{N(1-r)}{2N-1}$, $N \geq 2$
Thm. 6.1	$N = 4$, $d_i = \frac{2^i-1}{15}$, $r = \frac{1}{2}$, $\kappa = \frac{2}{5}$
Thm. 6.2	$d_i = \frac{i}{4}$, $\rho = \frac{1}{2} + \frac{1}{40}\binom{x}{2}$, whose c is not LC
Thm. 7.1, Cor. 7.2	$d_i = \frac{i}{N}$, $\rho = r(1+\varepsilon)^x$
Thm. 7.5	$\rho = r(1+\varepsilon)^x$ on every LC profile d
Thm. 7.6, Cor. 7.7	cubic ρ with $\sigma > 0$ and $t^2 < r\tau$ or $\tau^2 < t\sigma$ (Cor.: also $t\tau < r\sigma \leq \frac{7}{4}t\tau$)

9. Limits of the criterion

No uniform margin. A criterion valid for all N cannot leave a margin of fixed size around the Krawtchouk arrays. Let $\rho_N(r)$ be the supremum of the radii ρ such that every array satisfying (2.1), with $b_i \geq 0$ and with coefficients that differ from those of $K_{N,r}$ by at most ρ, has property (P). Raising the top mass of the orthogonality measure of $K_{N,r}$ (Proposition A.2) gives $\rho_N(r) \leq 3(1-r^2)^N$ whenever $(1-r^2)^N \leq \frac{1}{2}$ and $3(1-r^2)^N \leq (1-r)/N$ (Corollary A.3); both are proved in Appendix A. So, for every fixed $r \in (0,1)$, no set of arrays with property (P) contains coefficient balls of a radius independent of N around all $K_{N,r}$, and the log-concavity condition places these arrays on its edge. This is an upper bound for fixed r and $N \to \infty$ only. Lower bounds, the exact order $(1-r)^N$ for fixed N as $r \to 1$, and the constraints at $r = 1$ are proved in [17].

How sharp is log-concavity? The Krawtchouk arrays, with only $\nu_1 \neq 0$, and the arrays $G_{N,q,r}$, with equality in every log-concavity inequality, both lie on the boundary of the hypothesis of Theorem 3.1. Positivity of the Newton coefficients of the profile does not suffice. For $q > 1$ and $0 < \lambda < 1$ the combination $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ is a stochastic array with constant ratio b_i/c_i and positive Newton coefficients, and $\nu_2^2 < \nu_1\nu_3$ for $N \geq 3$ (Section 3). Exact computation, machine-verified in Lean, gives $g(N, N; N) < 0$ for seven such combinations, with r between $2/7$ and $4/5$ (Appendix B). At these parameters no sufficient criterion whose admitted set is convex, such as a system of linear inequalities in the coefficients, can admit both $K_{N,r}$ and $G_{N,q,r}$.

Log-concavity is not necessary either. For $0 < \mu < 1$ and $q > 1$ let $D(i) = (1-\mu)i + \mu[i]_q$, whose Newton coefficients are $\nu_1 = 1$ and $\nu_k = \mu(q-1)^{k-1}$ for $k \geq 2$; thus $\nu_1\nu_3/\nu_2^2 = 1/\mu$, and the other log-concavity inequalities hold with equality. For $\mu = \frac{1}{10}$ and $q = \frac{11}{10}$, the constant-ratio arrays with profile $d_i = D(i)/D(N)$ have property (P) for every $r \in (0,1]$ and $N \in \{6, 8, 10, 12\}$ (machine-verified), whereas $\mu = \frac{1}{100}$, the same q and $N = 6$ give $g(6, 6; 6) < 0$ at $r = 1$. Let

$$V_N = \inf\{\mu^{-1} : 0 < \mu < 1,\ q > 1,\ \text{the array with profile } D/D(N) \text{ and } r = 1 \text{ fails (P)}\},$$

with $\inf \varnothing = \infty$. Numerical searches for the smallest failing $1/\mu$ are recorded in Appendix B.

10. Open problems

The exact boundary. Determine the boundary between log-concavity and positivity of the Newton coefficients of the profile: the set of profiles for which every constant-ratio array has property (P). In particular, decide whether the threshold V_N of Section 9 tends to 1 as $N \to \infty$. The examples of Section 9 suggest that when log-concavity fails, the failures of (P) are driven by an acceleration of the increments near the terminal index.

The stability radius. For fixed $0 < r < 1$, determine the order of $\rho_N(r)$ as $N \to \infty$, and in particular decide whether $\liminf_N \rho_N(r)^{1/N}$ is positive. Corollary A.3 gives $\limsup_N \rho_N(r)^{1/N} \leq 1 - r^2$.

Other ratios. The criteria of Sections 5–7 cover several increasing ratios, but simple shape restrictions alone do not imply property (P). Exact counterexamples show that log-concavity of the Newton coefficients of both d and c with nondecreasing b, a merely increasing ratio with nondecreasing b, and nonnegative Newton coefficients of a cubic ratio can each fail. The parameters and negative product coefficients are collected in Appendix C, separately from sampled tests, which establish no threshold.

Find direct inequalities for other varying ratios, especially those outside Theorems 5.1, 6.1 and 7.5. Determine whether the cubic bound of Corollary 7.7 can be enlarged, and whether the log-concavity assumption on ratios of degree greater than three can be weakened to conditions like the one in Theorem 7.6. The coefficientwise detour test fails on geometric profiles beyond the factor 2 (Remark 7.8), which is a limit of that test rather than a counterexample to property (P). Find a proof of the key inequality (7.2) that does not use computer certificates. All these questions include the terminal row and products whose degree sum exceeds N.

Monotonicity-type criteria. Szwarc's Theorems 3.2 and 3.3 in [25] are inequality conditions of monotonicity type for infinite systems. By Theorem 4.1, any such criterion for nonsymmetric finite systems, valid including the terminal indices, gives one for symmetric finite systems. Inequality conditions of this type for all products except those whose pairwise index sums all exceed N are given in [17]; for the remaining, deep products, Theorems 3.1 and 7.5 and the other criteria of Sections 5–7 give shape conditions that control them.

A. The upper bound for the stability radius

This appendix proves Corollary A.3. Let

$$\operatorname{dist}(\widetilde{a},\widetilde{b},\widetilde{c}) \;=\; \max_{0\le i\le N} \max\Bigl(\Bigl|\widetilde{a}_i - \frac{N-i}{N}\Bigr|, \Bigl|\widetilde{b}_i - \frac{(1-r)i}{N}\Bigr|, \Bigl|\widetilde{c}_i - \frac{ri}{N}\Bigr|\Bigr)$$

be the coefficient distance to $K_{N,r}$.

Lemma A.1 *(Corner value).* *For $0 < r \le 1$, the array $K_{N,r}$ has $g(N,N;N) = (1-r)^N$.*

Proof. Use the model in the proof of Corollary 3.6, with $p_0 = r/(1+r)$. There $R_N = K_N = Y_1 \cdots Y_N$ and $K_k = \binom{N}{k}^{-1} \sum_{|S|=k} Y_S$, where $Y_S = \prod_{l\in S} Y_l$. Since $Y_l^2 = (1-r)Y_l + r$,

$$R_N^2 \;=\; \prod_{l=1}^{N} \bigl((1-r)Y_l + r\bigr) \;=\; \sum_{S} (1-r)^{|S|} r^{N-|S|} Y_S.$$

Both sides of $R_N^2 = \sum_k g(N,N;k) R_k$ depend only on the number j of negative coordinates, and the identity holds on the support, that is, for every j; so it holds as an identity of functions of $(Y_1,\dots,Y_N)$. The 2^N monomials Y_S are orthogonal for the product measure, with $\mathbb{E}[Y_S^2] = r^{|S|} > 0$, hence linearly independent. The monomial $Y_{\{1,\dots,N\}}$ occurs only in K_N, with coefficient one, which gives $g(N,N;N) = (1-r)^N$. ∎

Thus the corner coefficient vanishes at $r = 1$. The next proposition raises the top mass of the orthogonality measure.

Proposition A.2 *(Raising the top mass).* *Let $N \geq 1$ and $0 < r < 1$. The measure of $K_{N,r}$ has nodes $x_n = 1 - (1+r)n/N$ and masses $w_n = \binom{N}{n} r^{N-n}/(1+r)^N$. For $\tau > 0$ let μ^τ be this measure with the mass at $x_0 = 1$ raised to $(1+\tau)w_0$. Then μ^τ has*

$$g(N, N; N) = \frac{(1+\tau)^2(I - w_0) + w_0}{(1+\tau)(h - w_0) + w_0}, \qquad I = \big(r(1-r)\big)^N, \quad h = r^N,$$

which is negative exactly when $\tau > \tau_N^* = \big(1 - (1-r^2)^N\big)^{-1/2} - 1$. After the affine normalization $R_n(1) = 1$, $R_1 = x$, its recurrence coefficients differ from those of $K_{N,r}$ by at most $3\tau \max_n w_n$.

Proof. Write π_n and $S_n = \sum_{j<n} \pi_j$ for $K_{N,r}$, and put $\varepsilon = \tau w_0$. The polynomial $(1 + \varepsilon S_n)R_n - \varepsilon \sum_{j<n} \pi_j R_j$ has degree n and equals 1 at x_0. For $\deg p < n$, its integral against p with respect to $\mu^\tau = \mu + \varepsilon\delta_{x_0}$ is $-\varepsilon p(x_0) + \varepsilon p(x_0) = 0$, because $\sum_{j<n} \pi_j R_j$ reproduces $p(x_0)$. Hence it is the polynomial R_n^τ of μ^τ.

On the support, dual orthogonality gives $\sum_{j \leq N} \pi_j R_j(x) = \mathbf{1}_{x=x_0}/w_0$, and in particular $S_N + \pi_N = 1/w_0$. Therefore $R_N^\tau = (1+\tau)R_N - \tau \mathbf{1}_{x_0}$ on the support. By Lemma A.1 and $\pi_N = r^{-N}$ (Lemma 2.1), $\int R_N^3 \, d\mu = I$; also $\int R_N^2 \, d\mu = h$. Hence

$$\begin{aligned} \int (R_N^\tau)^3 d\mu^\tau &= (1+\tau)^3(I - w_0) + (1+\tau)w_0, \\ \int (R_N^\tau)^2 d\mu^\tau &= (1+\tau)^2(h - w_0) + (1+\tau)w_0, \end{aligned}$$

and their ratio is the stated formula. Since $r > r/(1+r) > r(1-r)$, we have $h > w_0 > I$. So the denominator is positive, and the numerator is negative exactly when $(1+\tau)^2 > w_0/(w_0 - I) = \big(1 - (1-r^2)^N\big)^{-1}$.

For the coefficients, comparing leading coefficients and using detailed balance, $\pi_n a_n = \pi_{n+1} c_{n+1}$, for both measures gives

$$a_n^\tau = a_n \frac{1 + \varepsilon S_n}{1 + \varepsilon S_{n+1}}, \qquad c_n^\tau = c_n \frac{1 + \varepsilon S_{n+1}}{1 + \varepsilon S_n},$$

and the row sums stay 1. Since $\varepsilon \pi_n = \tau w_n$, each of a_n, b_n, c_n changes by at most τw_n. Put $A_n = 1 + \varepsilon S_n$ and $t_n = \varepsilon \pi_n = \tau w_n$. Since $b_0^\tau = \varepsilon/(1+\varepsilon)$, the affine coordinate is $y = (1+\varepsilon)x - \varepsilon$, and the normalized coefficients are $\widetilde{a}_n = (1+\varepsilon)a_n^\tau$, $\widetilde{b}_n = (1+\varepsilon)b_n^\tau - \varepsilon$, $\widetilde{c}_n = (1+\varepsilon)c_n^\tau$. For $n \geq 1$ we have $A_n \geq 1 + \varepsilon$, and

$$\begin{aligned} \widetilde{a}_n - a_n &= \varepsilon a_n - (1+\varepsilon) a_n t_n / A_{n+1}, \\ \widetilde{c}_n - c_n &= \varepsilon c_n + (1+\varepsilon) c_n t_n / A_n, \\ \widetilde{b}_n - b_n &= -\varepsilon(a_n + c_n) + (1+\varepsilon) t_n \big(a_n/A_{n+1} - c_n/A_n\big). \end{aligned}$$

Because $a_n, c_n \geq 0$ and $a_n + c_n \leq 1$, each absolute change is at most $\varepsilon + t_n \leq 2\tau \max_j w_j$. Row zero is restored exactly. This proves, in particular, the stated bound $3\tau \max_j w_j$ for every $\tau > 0$. ∎

Corollary A.3 *(Exponential upper bound for the stability radius).* *Let $0 < r < 1$, and let $\rho_N(r)$ be the supremum of the ρ such that every array satisfying (2.1), with $\widetilde{b}_i \geq 0$ for all i and with $\operatorname{dist}(\widetilde{a}, \widetilde{b}, \widetilde{c}) \leq \rho$, has property (P). If $(1-r^2)^N \leq \frac{1}{2}$ and $3(1-r^2)^N \leq (1-r)/N$, then*

$$\rho_N(r) \leq 3\tau_N^* \max_n w_n \leq 3(1-r^2)^N.$$

In particular $\rho_N(r) = O\big((1-r^2)^N\big)$ as $N \to \infty$, for every fixed $r \in (0,1)$.

Proof. Put $x = (1-r^2)^N$ and $W = \max_n w_n$. For $0 < x \leq \frac{1}{2}$,

$$(1+x)^2(1-x) - 1 = x(1-x-x^2) > 0,$$

so $\tau_N^* < x$. Choose $\tau \in (\tau_N^*, x)$. Proposition A.2 gives a negative corner coefficient and full-array distance at most $3\tau W < 3x \leq (1-r)/N$. Thus $\widetilde{b}_n > 0$ for $n \geq 1$, while $\widetilde{b}_0 = 0$. The exact coefficient formulas and the positive affine scale give $\widetilde{a}_n > 0$ for $n < N$, $\widetilde{c}_n > 0$ for $n > 0$, and all the basic boundary and row-sum equalities. Letting $\tau \downarrow \tau_N^*$ proves $\rho_N(r) \leq 3\tau_N^* W \leq 3x$. For each fixed $r \in (0,1)$, both smallness conditions hold for all sufficiently large N, since $N(1-r^2)^N \to 0$. ∎

Thus, for every fixed $r \in (0,1)$, no set of arrays with property (P) contains coefficient balls of a positive radius independent of N around all $K_{N,r}$. Corollary A.3 bounds $\rho_N(r)$ from above only; it does not determine its order. For each fixed N, $\rho_N(r) > 0$ for $0 < r < 1$: explicit perturbation inequalities give $\rho_N(r) \geq \frac{1}{2}(\min(r, 1-r)/(32N))^N$, a lower bound that decays faster than every exponential in N, and, for fixed $N \geq 2$, $\rho_N(r)$ is of exact order $(1-r)^N$ as $r \to 1$; at the symmetric endpoint $r = 1$, criteria that admit polynomial deformation rules, or analytic rules of the kind specified there, must satisfy two first-order equalities. These results are proved in [17].

B. Examples and searches for Section 9

Convex combinations. The combination $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ has increments proportional to $1 + t\,q^{i-N}$, where $t = \lambda N(q-1)q^{N-1}/((1-\lambda)(q^N-1))$. Exact computation, machine-verified in Lean, gives $g(N, N; N) < 0$ at the parameters

$$(r, N, q, t) = (4/5, 10, 10, 1),\ (3/5, 20, 10, 2),\ (1/2, 20, 10, 1),\ (2/5, 60, 10, 2),$$
$$(1/3, 100, 10, 2),\ (3/10, 200, 2, 5),\ (2/7, 200, 10, 5),$$

while $K_{N,r}$ and $G_{N,q,r}$ have property (P) by Corollaries 3.6 and 3.7.

The profiles $D/D(N)$. For $\mu = \frac{1}{10}$ and $q = \frac{11}{10}$, the exact computation shows that every coefficient $g(m, n; k)$, $N \in \{6, 8, 10, 12\}$, is a polynomial in r with nonnegative Bernstein coefficients, which gives property (P) for every $r \in (0, 1]$.

Searches. At $r = 1$ we searched coarse grids of μ and q, with float64 screening and high-precision confirmation for $N \le 20$ and high-precision arithmetic for $20 \le N \le 80$. The smallest factor $1/\mu$ for which (P) failed was $10^{5/2}$ at $N = 4$, 10^2 for $6 \le N \le 30$, and $10^{3/2}$ at $N = 40, 60, 80$; scripts and outputs are in the companion code [18]. These are the smallest failures found, not proven thresholds: the observed failures lay in narrow windows of (μ, q) on the tested grids, and none of these searches is formalized.

C. Exact counterexamples and sampled tests

All examples in this appendix are reproduced in the companion code [18]. The counterexamples of the first paragraph are exact and machine-verified. The tests of the second paragraph are sampled numerical evidence; they are not formalized.

Exact counterexamples. Each array below satisfies (2.1) and has a negative product coefficient.

- *Log-concave Newton coefficients of d and c with nondecreasing b.* The arrays $d = (0, 25, 55, 91)/91$ and $c = (0, 24, 54, 90)/91$ give $g(3, 3; 3) = -3/2002$.

- *An increasing concave ratio with nondecreasing b.* The linear profile $d_i = i/3$ with $\rho = (1/2, 3/4, 5/6)$, for which $b_i = 1/6$ for $i \ge 1$, gives $g(3, 3; 3) = -1/24$.

- *A cubic ratio with nonnegative Newton coefficients.* For $N = 4$, $d_i = i/4$ and $\rho(x) = \frac{19}{20} + \binom{x}{3}/320$, whose Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$ are nonnegative but have internal zeros, $b = (0, \frac{1}{80}, \frac{1}{40}, \frac{9}{256}, \frac{3}{80})$ is nondecreasing and $g(4, 4; 4) = -14011/30720000$. So Theorem 6.2 does not extend to cubic ratios without further conditions.

- *Cubic ratios with $r \le \frac{9}{10}$.* On linear profiles with $\kappa = 1/80$, which keeps b nondecreasing, $\rho_i = \frac{9}{10} + \kappa\binom{i}{3}/\binom{N}{3}$ gives $g(48, 48; 48) < 0$ for $N = 48$, and $\rho_i = \frac{9}{10} + \kappa(i/N)^3$ gives $g(64, 64; 63) < 0$ for $N = 64$.

- *Steep ratios.* On linear profiles, with nondecreasing b and nonnegative Newton coefficients of ρ, the ratio $\rho_i = \frac{1}{2} + 10^{i-N}/100$ gives $g(40, 40; 40) < 0$ for $N = 40$. Raising only the last ratio, $\rho_i = r$ for $i < N$ and $\rho_N = r + (1 - r)/N$, gives $g(40, 40; 38) < 0$ for $r = \frac{1}{2}$, $N = 40$, and $g(120, 120; 119) < 0$ for $r = \frac{2}{5}$, $N = 120$; these two ratios are convex, with a single kink.

Write a cubic ratio as $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The cubic examples violate $r\sigma \le t\tau$, a consequence of log-concavity without internal zeros of (r, t, τ, σ): in the first two $t = \tau = 0$, and for $\kappa(i/N)^3$ one has $t\tau = 6\kappa^2/N^6 < r\sigma = 6r\kappa/N^3$. One further exact computation concerns the method rather than property (P): for $t = \tau = 0$ the operator M of the proof of Theorem 6.1 can have negative entries, for instance for $N = 6$, $d_i = i/6$, $r = \frac{1}{5}$ and $\sigma = \frac{1}{200}$. This obstructs extending that resummation to such data, and does not contradict Theorem 6.1, which does not cover them.

Sampled tests without failures. These are numerical evidence over finitely many arrays, not certified thresholds. In sampled arrays with $t, \tau \geq 0$, $r\sigma = u\,t\tau$ and nondecreasing b on log-concave Newton profiles, no negative coefficients were found for $u \in \{\frac{1}{4}, 1, 4, 16, 64\}$, $r \in \{\frac{1}{10}, \frac{1}{3}, \frac{1}{2}, \frac{4}{5}, \frac{19}{20}, \frac{99}{100}\}$ and $4 \leq N \leq 18$ (full tables), or in the coefficients $g(m, N; k)$ for $u \in \{1, 4, 16\}$ and $N = 24, 32, 48, 64, 96$. In these samples the operator M of the proof of Theorem 6.1 was nonnegative for $u \leq 4$ and $N \leq 13$. The samples include values $u > \frac{7}{4}$, so they bear on the question of Section 10 whether the bound of Corollary 7.7 can be enlarged, but they do not decide it.

D. Proof of Proposition 7.3

We recall the statement: if $(\rho_i)_{i\geq 0}$ and $(\nu_j)_{j\geq 0}$ are nonnegative and log-concave without internal zeros, $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$, then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of fg are nonnegative and log-concave without internal zeros.

Proof. The product rule $\binom{x}{i}\binom{x}{j} = \sum_k \binom{k}{i}\binom{i}{k-j}\binom{x}{k}$ gives $\kappa_k = \sum_p H_k(p)\rho_{p_1}\nu_{p_2}$, where $H_k(i,j) = \binom{k}{i}\binom{i}{k-j} = [y^i w^j](y + w + yw)^k$ vanishes unless $i, j \leq k \leq i + j$. If either sequence vanishes identically, so does κ. Otherwise, if ρ and ν are supported on $[p_0, p_1]$ and $[q_0, q_1]$, with $p_1, q_1 \leq \infty$, then $\kappa_k > 0$ exactly for $\max(p_0, q_0) \leq k \leq p_1 + q_1$, so κ has no internal zeros. For log-concavity fix $k \geq 1$; then

$$\kappa_k^2 - \kappa_{k-1}\kappa_{k+1} = \sum_{c\in\mathbb{Z}^2} \sum_{p\in\mathbb{Z}^2} T_c(p)\, u_c(p_1)\, w_c(p_2),$$
$$T_c(p) = H_k(p)H_k(c-p) - H_{k-1}(p)H_{k+1}(c-p),$$

with $u_c(i) = \rho_i\rho_{c_1-i}$ and $w_c(j) = \nu_j\nu_{c_2-j}$. By log-concavity without internal zeros, u_c is symmetric about $c_1/2$ and nondecreasing on $i \leq c_1/2$, so $u_c = \sum_{0\leq a\leq c_1/2}(u_c(a) - u_c(a-1))\mathbf{1}_{[a,c_1-a]}$ with nonnegative coefficients, and likewise w_c. It therefore suffices to show $D_c(B) = \sum_{p\in B} T_c(p) \geq 0$ for every box $B = [a, c_1 - a] \times [b, c_2 - b]$ with $a \leq c_1/2$, $b \leq c_2/2$.

Let $E = \{(1,0), (0,1), (1,1)\}$, so that $H_k = \sum_{e\in E} H_{k-1}(\cdot - e)$. Expanding $H_k(p)$, $H_k(c-p)$ and $H_{k+1}(c-p)$ in this way gives

$$D_c(B) = \sum_{e,e'\in E} \big(V_{e+e'}(B-e) - V_{e+e'}(B)\big), \qquad V_s(p) = H_{k-1}(p)H_{k-1}(c-s-p),$$

where $V_s(X) = \sum_{p\in X} V_s(p)$. Fix $s = e + e'$, write $V = V_s$, $m = c - s$, $t = k - 1$ and $\mu = m/2$. With $n_1 = 2t - m_2$, $n_2 = 2t - m_1$, $n_3 = m_1 + m_2 - 2t$ one has $V(x,y) = \frac{t!^2}{n_1!n_2!n_3!}\binom{n_1}{t-y}\binom{n_2}{t-x}\binom{n_3}{x+y-t}$ (and $V = 0$ if some $n_i < 0$). Three properties follow.

(T1) For every interval W, the marginal $G_W(x) = \sum_{y\in W} V(x,y)$ is a constant times $\binom{n_2}{t-x}\sum_{z\in t-W}\binom{n_1}{z}\binom{n_3}{x-z}$, a product of a binomial row and a convolution of two log-concave sequences without internal zeros; so G_W is log-concave without internal zeros. The same holds with the roles of the coordinates exchanged.

(T2) $V(c-s-p) = V(p)$, and V is reverse regular of order two: $V(x,y)V(x',y') \le V(x,y')V(x',y)$ for $x \le x'$, $y \le y'$, by log-concavity of $\binom{n_3}{\cdot}$. Taking $x' = 2\mu_1 - x$ and y, y' symmetric about μ_2, central symmetry turns this into $V(x,y)^2 \le V(x,y')^2$. Hence $V(x,y') \ge V(x,y)$ whenever $x \le \mu_1$, $y \le y'$ and $y + y' = 2\mu_2$; and symmetrically in the coordinates.

(T3) If G is log-concave without internal zeros and $G(x) \le G(2\mu_1 - x)$ for all $x \le \mu_1$, then G is nondecreasing on $x \le \mu_1$: otherwise $G(x-1) > G(x)$ for some $x \le \mu_1$, and log-concavity makes G nonincreasing from $x-1$ on, so $G(2\mu_1 - x + 1) \le G(x) < G(x-1)$.

Write $I = [a, c_1 - a]$, $J = [b, c_2 - b]$, $X(I', J') = V(I' \times J')$, and telescope along rows or columns. By the symmetry $H_k(i,j) = H_k(j,i)$ it suffices to treat five ordered pairs.

$e = e' = (1,0)$: $\mu = c/2 - (1,0)$ and $V(B-e) - V(B) = G_J(a-1) - G_J(a-2)$ by central symmetry. Since J is symmetric about μ_2, G_J is symmetric about μ_1, hence nondecreasing on $x \le \mu_1$ by (T1) and (T3); and $a - 1 \le \mu_1$.

$e = (1,0)$, $e' = (0,1)$: $\mu = (c - (1,1))/2$ and $V(B-e) - V(B) = V(a-1, c_2 - b) - V(a-1, b-1) \ge 0$ by (T2), since $a - 1 < \mu_1$ and $(c_2 - b) + (b-1) = 2\mu_2$.

$e = (1,0)$, $e' = (1,1)$: $\mu = c/2 - (1, \frac{1}{2})$ and

$$V(B-e) - V(B) = \big[V(a-1, c_2 - b) - V(a-1, b-1)\big] + \big[G_{J-1}(a-1) - G_{J-1}(a-2)\big].$$

The first bracket is nonnegative by (T2), as $a - 1 \le \mu_1$. For the second, $G_{J-1}(2\mu_1 - x) - G_{J-1}(x) = V(x, c_2 - b) - V(x, b-1) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) apply.

$e = (1,1)$, $e' = (1,0)$: the same V, and $V(B-e) - V(B) = \big[X(I-1, J-1) - X(I-1, J)\big] + \big[V(B - (1,0)) - V(B)\big]$. The first bracket vanishes by central symmetry, since $I - 1$ is symmetric about μ_1; the second is the previous case.

$e = e' = (1,1)$: $\mu = c/2 - (1,1)$ and

$$\begin{aligned} V(B-e) - V(B) = {} & \big[K(b-1) - K(b-2)\big] + \big[X(a-1, J) - X(a-1, J-2)\big] \\ & + \big[G_{J-2}(a-1) - G_{J-2}(a-2)\big] \end{aligned}$$

with $K(y) = \sum_{x \in I-1} V(x,y)$. Since $I - 1$ is symmetric about μ_1, K is symmetric about μ_2 and nondecreasing on $y \le \mu_2$ by (T1) and (T3); and $b - 1 \le \mu_2$. The second bracket is $[V(a-1, c_2 - b) - V(a-1, b-2)] + [V(a-1, c_2 - b - 1) - V(a-1, b-1)]$, two differences of the form in (T2), since both pairs of second coordinates sum to $2\mu_2 = c_2 - 2$. For the third, $G_{J-2}(2\mu_1 - x) - G_{J-2}(x) = X(x, J) - X(x, J-2) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) give $G_{J-2}(a-1) \ge G_{J-2}(a-2)$.

Each of the nine ordered-step contributions $V_{e+e'}(B-e) - V_{e+e'}(B)$ is nonnegative, so $D_c(B) \ge 0$, which proves the proposition. (The individual $T_c(p)$ need not be nonnegative.) ■

Diagnostics in the companion code [18] check in exact arithmetic the identity for $D_c(B)$, the sign of each of its nine terms for $k \le 9$, and every bracket of the case analysis, (T1) and (T2) for $k \le 6$.

AI-use disclosure

Claude Opus 5.5 and, for parts of the Lean formalization of Theorem 7.5, Claude Haiku (Anthropic; the Haiku version was not recorded), both run in the Claude Code agent; GPT-6 Astra or GPT-6.1 Sol (OpenAI; which of the two was not recorded), run in the Codex agent; and GPT-6 Pro (OpenAI), used in ChatGPT, were used to develop the proofs, the exact checks and certificates, the Lean formalization and the text. GPT-6 Pro also reviewed every result. This AI review is not human peer review, and no independent human verification is claimed. The author set the research objective and is responsible for the mathematical claims. The supplements [20, 21] give the full disclosure, including which results arose with which model.

Code and formalization

All theorems, corollaries and propositions, and Lemmas 2.1, A.1 and 7.4, are formalized in Lean 4 with Mathlib in the companion code [18], whose axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`; "machine-verified" refers only to these Lean theorems. The certificate scripts were also run unmodified from a fresh clone at the cited revision [19]. The lemmas of Section 3 are formalized within the proof of Theorem 3.1, whose Lean proof follows the written one.

Some Lean proofs take another route, through the Lean development of [17]. Lean derives the positivity statements of Corollaries 3.6, 3.7 and 4.2, and through Corollary 4.2 that of Theorem 4.3, from the geometric-increment criterion of [17] rather than from Theorem 3.1. It proves Lemma A.1, on which Proposition A.2 and Corollary A.3 rest, in the algebra of [17]. These Lean proofs verify the statements given here by routes different from the written proofs. The written proofs do not use [17]; its Lean development uses this paper's for two proofs.

This paper merges two earlier versions, arXiv:2611.01561v1 (Sections 2–4 and 9, Appendices A and B) and the companion paper *Increasing ratios in finite nonnegative linearization* of release 1.0.6 of [18] (Sections 5–8, Appendices C and D). The supplements [20, 21], written for these versions and cited in their numbering, list the Lean declaration for each result and remark, describe the Lean routes that differ from the written proofs, give the inventory of computer-assisted steps with their ranges, scripts and Lean declarations, and list what is not formalized: the comparisons with the literature, the numerical searches concerning V_N and the sampled tests of Appendix C. The documentation [19] describes the exact-arithmetic diagnostics.

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